ADM Reverse Engineering Study

Final Report

Team 05

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Chapter 1

Mission Goals and Requirements

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1.1 Mission high level goals

The ADM-Aeolus mission was designed to demonstrate the effectiveness of the DWL technique in providing near-real-time data on the wind profile within 30 km from Earth's surface. This constitutes the main goal of the mission.
To demonstrate the correct functioning of the payload, the data obtained from the Lidar LOS are subsequently fed into NWP models, in order to improve 3D analysis and forecasting of the wind vector field.

In addition, the mission was aimed at accomplishing some secondary objectives:

- providing data sets for the evaluation of climate models and more generally using wind profile data for climate research;
- retrieving measurements about the optical properties of clouds and aerosol particulate.

The mission would therefore provide essential data to tackle key issues identified by the WCRP, such as measuring climate variability, assessing and refining climate models, and conducting process studies related to climate change.

In such scientific context, the choice of a satellite-based DWL is justified, despite its cost, complexity and experimental technology, thanks to its ability to measure wind velocity directly and globally. In fact, other methods for wind velocity determination exploit radiosondes and weather radars on board airliners, which offer local information, as well as indirect measurement through pressure and temperature.

1.2 Payload functioning and mission drivers

The mission drivers are based on the payload's architecture and operations; therefore, in the following paragraph, the main operating characteristics of the latter will be introduced, so that the drivers will eventually derive from these in a natural way.

The Aeolus spacecraft carries a DWL called ALADIN, which is a pulsed UV Lidar used to recover the wind speed exploiting the Doppler effect. The measurement technique is based on the frequency shift of emitted laser light, which is backscattered by air molecules or aerosol cloud particles.
Since the atmosphere is composed of air molecules, aerosol and water vapour, the wavelength at which the measurements are provided must vary to allow a complete mapping of the wind profile. Rayleigh backscattering is inversely proportional to the fourth power of the wavelength of light. The main consequence of this is that it must be used at short wavelengths and therefore it is effective for measuring the wind in the presence of air molecules, whose sizes are smaller than the wavelength itself. Conversely, Mie backscattering works at a wavelength comparable to the size of aerosol and cloud droplets; therefore, it is effective for measuring the wind in layers at the top of optically thick clouds. Thus, it is evident that both Mie and Rayleigh backscattering need to be exploited at the same time by the spacecraft [1, Sec. 8.1]. The shift is proportional to the relative motion of the scatterers along the LOS direction of the laser, through which the parallel velocity is found.

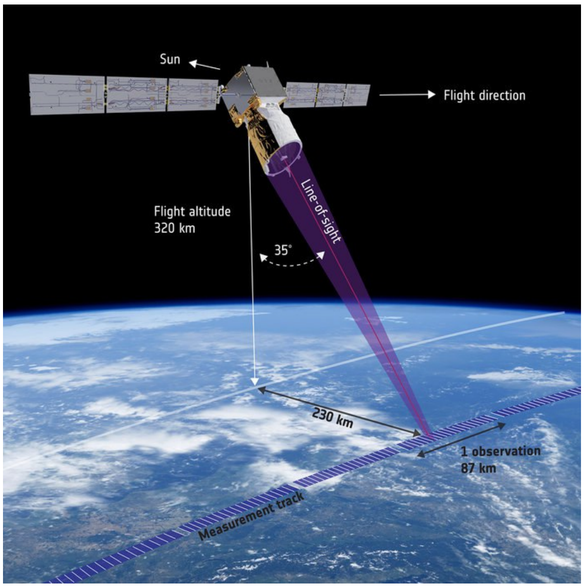
The wind velocity along the LOS is used to retrieve the component of wind speed that lies on the horizontal plane, as a projection by means of the angle between the LOS and the vertical direction. This component is the one needed by the NWP models to perform climate predictions; consequently, the angle between the vertical and the LOS needs to be known very precisely.

Given the importance of having a precise knowledge of the angle, it appears clear that the pointing and stability requirements are crucial for the optimal fulfilment of the mission. Having a major impact on the design and the cost of the spacecraft, such aspects become drivers.

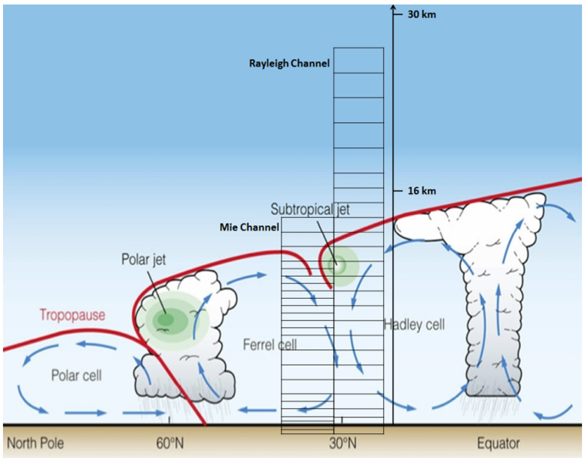
The model relies on the component of the velocity projected on the horizontal plane because it represents a compromise between the complexity arising from a three-dimensional vector field and an approximately correct representation of the movement of air masses in the low atmosphere, at a global level. The horizontal component of the wind is significant from the point of view of the representation of winds worldwide since the vertical component is of an order of magnitude lower than the one used; although the vertical one is instead essential for the representation at a local scale [2].

Furthermore, the satellite needs to transmit data in very short time, hence it is necessary for it to be equipped with adequate instrumentation to undertake this task. From the mission requirements document [1], it emerges that the satellite must provide a horizontal spatial resolution of less than 100 km, while maintaining a sub-sampling of 3 km, which, given the speed of the satellite, would correspond to about 2 sub-samplings per second [1, Sec. 8.3.2]. This is due to the need for constant measurement to build a more detailed wind profile. In addition, the same document reports that the sampling rates shall be sufficiently high to avoid data loss caused by under-sampling [3, Sec. 6.9.2].

Given these considerations, it is clear that the instruments implemented must have certain specifics to provide a high data rate, thus constituting the second mission driver.



(a) Aeolus measurement geometry



(b) Rayleigh and Mie channels

1.3 Functional analysis

Aeolus mission's functional analysis, describing the logical sequence of essential functionalities to be performed during the mission, is shown in Figure 1.2.

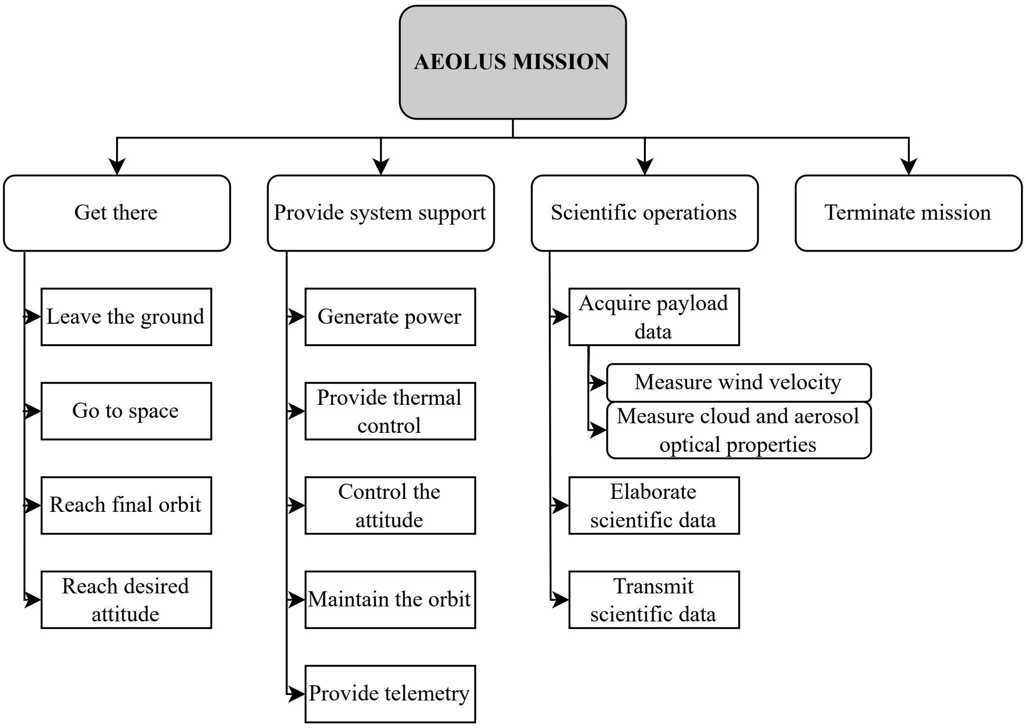


Figure 1.2: Functional analysis

1.4 Mission phases and ConOps

The main phases that characterize the Aeolus mission are: **LEOP, Commissioning, Operational, End of Life.** They are placed in time as displayed in Figure 1.3.

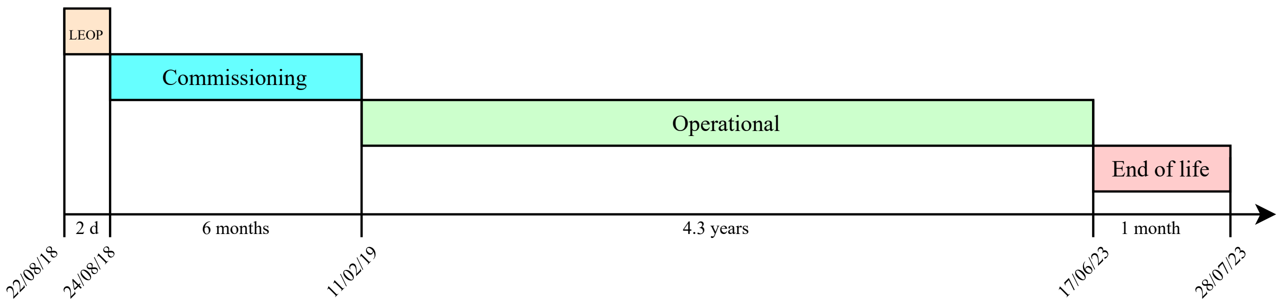


Figure 1.3: Mission phases

By analysing the history of the mission [4], Table 1.1 was made, where the phases are linked to the functionalities through the related operations.

Phase	Conceptual Operations	Functions
LEOP	Launch Separation from launcher	Leave the ground Go to space Reach final orbit
	Detumbling Slew Manoeuvre	Reach desired attitude
	Deployment of solar panels Test onboard systems Point the antenna Establish link with ground	Provide system support
Commissioning	ALADIN pointing, activation and calibration Share scientific data with ground Validate the accuracy of measured data	Provide system support Scientific operations
Operational	Periodical health check-ups Orbit correction manoeuvres Collision avoidance manoeuvres Provide and store high quality wind data Improve data gathering Deliver forecasts to end users	Provide system support Scientific operations
End of life	Lower S/C altitude through manoeuvres Burn up in the atmosphere	Terminate mission

Table 1.1: Link between phases, conops and functionalities

The conceptual operations displayed in the table are proposed in chronological order from top to bottom. This is valid for all the phases, made exception for the operational phase, where the operations are periodically cycled through to guarantee nominal activities. The third column contains the functionalities satisfied by the conceptual operations. Moreover, the objective of Table 1.1 is to correlate the payload's functioning, by means of the conceptual operations, to the phases. Finally, it is noteworthy to underline that the conceptual operation "improve data gathering" is linked to the fact that, as the mission progressed, there was a continuous improvement in data quality, thanks also to the collaboration between various nations and private agencies.

1.5 Mission requirements

The high-level mission requirements are compliant with the scientific constraints presented in Table 1.2. The table provides the measurement accuracy requested for an improvement in NWP. In addition, other design criteria are related to the coverage and frequency of observations. A set of possible mission requirements is now presented, justified both in terms of mission design and reverse engineering.

	Ideal requirements			Breakthrough requirements			Threshold requirements		
	LT	HT	Strat	LT	HT	Strat	LT	HT	Strat
Vertical Resolution [km]	0.5	0.5	0.5	1	1	1	3	3	3
Horizontal Domain	global								
Horizontal Resolution [km]	15	15	15	100	100	100	500	500	500
Temporal Sampling [hour]	1	1	1	6	6	6	12	12	12
Accuracy (Component) [ms^{-1}]	1	1	1	3	3	3	5	8	5
Timeliness [hour]	0.1			0.5			6		

Table 1.2: WMO observation requirements for Global NWP for the horizontal component of wind profiles (WMO, 2011). LT, HT, Strat.

R-1: Wind velocity shall be determined with the requested accuracy in clear atmosphere, below optically thin clouds and on top of optically thick clouds.

Measuring accurate wind velocity profiles in a broad range of atmospheric conditions is mandatory for an improvement in NWP. In order to meet this requirement, ALADIN exploits Rayleigh and Mie backscattering, which are due to atmospheric air molecules and aerosol and cloud particles respectively.

R-2: The mission shall have global coverage.

This requirement can be inferred through the analysis of the satellite trajectory and is also compliant with the horizontal domain scientific requirement of Table 1.2.

R-3: The mission shall provide wind velocity measurements from Earth surface up to lower stratosphere.

Radiosondes are able to reach the altitude of 20 km [5]. Yet, they only provide local measurements. The acquisition of global wind velocity profiles at such altitude is beneficial because of the lack of data in upper troposphere and lower stratosphere.

R-4: The mission shall provide globally distributed measurements every 12 hours.

Since the data acquired by Aeolus are used for weather forecasting, their availability must be compatible with the timescale of NWP models. Such requirement can be understood by the analysis of the ground track covered by the satellite in the specified time, as it influenced the choice of the orbit altitude.

R-5: The mission shall provide at least 100 independent observations every hour.

As shown in Figure 1.1a, the satellite tilt of 35° with respect to the nadir direction allows to represent each wind observation profile as a box of approximately 200 km × 200 km, which is a good compromise between the threshold requirement and the ideal one set by the scientific community for the horizontal resolution in Table 1.2. The actual mission performances exceed the threshold value of 100 independent observations per hour set by WMO [1, Sec. 8.3.3.1].

R-6: The mission shall deliver wind observation data within 3 hours (near-real-time).

Medium-range weather forecast requires data recent enough for effective predictions. This requirement is therefore in accordance to what demanded in Table 1.2. The choice of a LEO near-polar orbit, allowed the satellite to communicate with the Svalbard ground station at each orbit (approximately every 90 minutes) which allowed short-time data transmission.

R-7: The mission shall last at least 3 years. The design should account for a possible increase in the duration of the mission.

The required mission duration was set in order for the instrument to be calibrated, its results validated and introduced in NWP models to demonstrate their positive impact. The mission was then extended due to its significant results [6]. It was stopped one year later because of increased disturbances due to solar activity and shortage of fuel, which was needed for the controlled re-entry.

R-8: The satellite shall provide the desired pointing during all mission phases.

This requirement is derived from one of the mission drivers presented in section 1.2. Moreover, the presence of an optical instrument requires its protection from Sun exposure, as well as precise pointing to minimize observation errors. Precise pointing is also aimed at minimizing satellite velocity components along the instrument's LOS. The satellite is provided with several attitude sensors and actuators whose presence is needed in a mission of such kind, despite their cost and complexity [7, Sec. 4.4.2].

1.6 Mission analysis

The main drivers for orbit selection are altitude and coverage. For an Earth-observation mission like Aeolus, the quality and resolution of acquired data is improved with a lower altitude. Therefore, a LEO is suitable for this purpose. On the other hand, the design should also account for increased aerodynamic drag resulting from the choice of a very low altitude orbit, which is linked to an increase in on-board fuel storage needed due to more frequent orbital corrections, thus higher costs. Therefore, the choice of a suitable altitude results from a trade-off between such contrasting requirements. The satellite is positioned in LEO, which allows the mission to meet the requirement of globally distributed measurements every 12 hours. The global coverage requirement is satisfied through the choice of a near-polar orbit, which is able to cover a broad latitude range.
The satellite is placed in a dawn-dusk SSO, which provides optimal environmental conditions both for scientific and operational purposes. In fact, during twilight, stable thermal conditions ensure lower number of clouds, thus improving wind observations. Also, the payload is always pointed in the direction opposite to the Sun, thus reducing background radiation. This solution also allows the maximisation of power available to solar arrays and constant thermal conditions on the satellite [7, Sec. 4.2.2].
The selected orbit has an average altitude of 320 km and an inclination of 96.97°. A repeating cycle of 111 full orbits over 7 days is designed, in order to organize the nominal mission operations according to a weekly schedule [4][8].
The satellite is injected by the launcher directly in the desired orbit [9], where it spends its whole operational life. Weekly correction manoeuvres allow it to keep the correct orbit. Even though it was not originally planned, the end-of-life phase included a controlled re-entry that allowed the satellite to burn up in the atmosphere, causing eventual remaining pieces to fall on Antarctica.

Chapter 2

Mission Analysis and Propulsion Subsystem

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2.1 Mission trajectory analysis

This paper examines the Aeolus mission's propulsion system, starting with an analysis of the overall mission trajectory and propellant consumption by means of delta velocities. As already stated in the previous report, the satellite spends its whole operational life in its nominal orbit. The main manoeuvres that the satellite shall be capable to perform are:

- **Launcher injection error correction**. The design shall account for the worst errors reported in the launcher datasheet [10, Sec. 2.5];
- **Orbit maintenance**. The satellite shall maintain an orbit suitable for the correct operation of its payload through periodic orbit corrections;
- **End of life**. The mission is required to end with the burning of the satellite into the atmosphere.

The total Δv used during the mission cannot be found from literature. Yet, it is possible to provide an estimate of the overall budget through the Tsiolkovsky equation from the satellite's total mass, the embarked propellant mass, and the features of its propulsive subsystem, which will be further discussed in the following section. The adopted values [11] are:

- Wet mass: $M_{wet} = 1366\,\mathrm{kg}$;
- Propellant mass: $M_{prop} = 266\,\mathrm{kg}$;
- Dry mass: $M_{dry} = 1100\,\mathrm{kg}$.

The resulting Δv budget is $\Delta v_{budget} = 480.4\,\mathrm{m\,s^{-1}}$. From system requirement AOS-20 [3, Sec. 6.6.4], the overall propellant mass includes a 20% margin with respect to that resulting from the design. Also, a margin of 5.5% embeds ullage, fuel loss and loading uncertainty. The value reported accounts for the above-mentioned manoeuvres and includes the required margins. The only real data available from literature concern the first two weeks of Station keeping (SK) and the manoeuvres for disposal. It is still possible to estimate the required costs for the manoeuvres through simplified models and applying the required margins, and compare it to the actual values.

2.1.1 Injection error correction

The satellite is to be injected into its nominal orbit by the Vega Launcher and, according to requirement AOS-1 [3, Sec. 6.6.1], it shall be able to correct the worst injection errors [10, Sec. 2.5]. The two most relevant contributions to the cost of such correction are the error in semi-major axis and inclination. Thus, the manoeuvre is modeled through a Hohmann transfer followed by a plane change. The values used in the calculations are $\Delta a = -15\,\mathrm{km}$ and $\Delta i = 0.15°$. According to the standards (MAR-DV-040 [12]), since the uncertainty is extended to 3σ, no additional margin on the manoeuvre Δv is applied.

2.1.2 Station keeping

Mission operations include orbit maintenance manoeuvres either weekly or every two weeks to counteract the effect of disturbances. In the present model, the orbit of the satellite is propagated for one week under the effect of J_2 and aerodynamic drag, the two most relevant perturbations in Low Earth Orbit (LEO). A Hohmann transfer is then performed to restore nominal conditions, providing an estimate for the average weekly Δv. The result is then multiplied by the nominal duration of the mission (~3.25 y) to obtain the overall cost of SK. The only effect considered by the model is the reduction of semi-major axis, while the change in inclination is neglected. Drag acceleration is modelled via the following expression

$$a_{drag} = \frac{1}{2}\rho v_{rel}^2 \frac{A_{cross}}{M} C_D \tag{2.1.1}$$

where $C_D = 2.2$, $M = M_{dry}$, $A_{cross} = 6.163\,\mathrm{m^2}$ and ρ is modelled considering that the satellite operates in a period of low solar activity [13]. Since a stochastic approach is adopted, a 100% margin is added to the computed value (MAR-DV-020 [12]).

2.1.3 End of life

For Aeolus's return, an assisted reentry is designed, which is an optimal choice to prevent the potential debris from landing in populated areas, given the remaining propellant reserves. The actual reentry strategy included the interruption of SK manoeuvres for approximately one month, to allow the satellite to decrease its altitude from 320 km to 275 km [14]. A sequence of seven manoeuvres, combined with the much stronger effect of drag, further lowered the orbit to an altitude of 150 km, after which the satellite fell and burned in the atmosphere.
The simplified model adopted reproduces the first free descent of the satellite due to drag accounting for medium solar activity, as displayed in Figure 2.1, showing good correspondence with the actual trajectory recovered from ephemerides [15][16], and merges the subsequent control impulses into a Hohmann transfer. According to the standards, a margin of $10\,\mathrm{m\,s^{-1}}$ is then added to the result (MAR-DV-010 [12]).

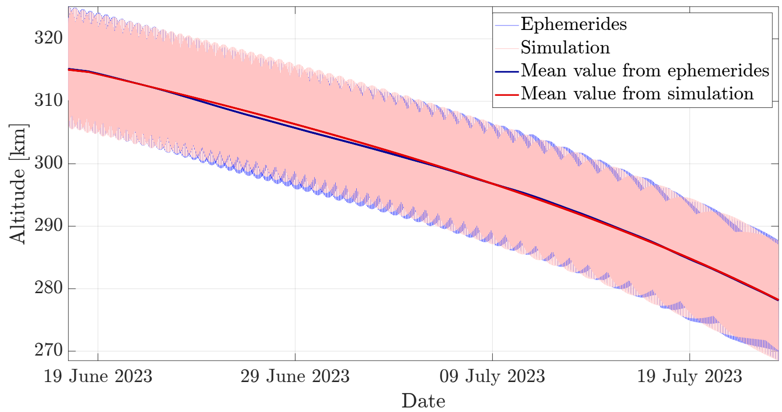


Figure 2.1: Altitude variation due to atmospheric drag between 2023/06/18 and 2023/07/24.

2.1.4 Δv budget breakdown

An estimate of the actual weekly cost of SK can be performed by averaging the data regarding the first weeks of the mission [13, Table 4]. The result is compared to the values obtained from the simplified models in Table 2.1.

Manoeuvre	**Real** $[\mathrm{m\,s^{-1}}]$	**Estimate** $[\mathrm{m\,s^{-1}}]$	
		Marginless	Margin
Injection error correction	N/A	28.87	28.87
SK (weekly average)	0.53	0.86	1.72
SK (overall)	N/A	145.92	291.83
Assisted reentry	59	75.74	85.74

Table 2.1: Δv expenditure for the main manoeuvres performed by the spacecraft.

It is relevant to note that both the cost of SK and the assisted reentry have been overestimated. The former because of the use of the dry mass in the computation, which increases the perturbing acceleration, the latter due to the overlooking of the contribution of drag during the Hohmann transfer.
Finally, it is possible to compare the estimated total Δv to that obtained removing the above-stated margins on the embarked propellant mass. The two amount, respectively, to $\Delta v_{estimate} = 406.44\,\mathrm{m\,s^{-1}}$ and $\Delta v_{design} = 394.40\,\mathrm{m\,s^{-1}}$. Again, the estimated value is higher than that obtained during mission design due to the previously reported approximations, even though the cost of attitude control manoeuvres and the de-saturation of reaction wheels has been neglected. However, the overall Δv_{budget} is much larger than that estimated, thus allowing not only the successful completion of the mission, but also its extension by over one year and the execution of exceptional manoeuvres such as one collision avoidance [17].

2.2 Propulsion subsystem

2.2.1 Introduction

The ADM-Aeolus mission is equipped with a monopropellant propulsion subsystem, featuring four thrusters with cold redundancy, used to undertake the main manoeuvres cited in section 2.1. It derives that the subsystem shall be capable of giving relatively small impulses to be able to balance out external perturbations and perform valid attitude manoeuvring.

2.2.2 Propulsion subsystem architecture

Due to the lack of data, it was not possible to determine the exact propulsion subsystem architecture of the ADM-Aeolus mission, thus the propulsion subsystem architecture of Copernicus Sentinel-2A, shown in Figure 2.2, is considered. In particular, the latter assumption is reasonable due to the fact that both of them share similar tanks [18], exploit the propulsion subsystem mainly for station keeping, use hydrazine as monopropellant and helium as pressurant and have similar masses. Additionally, both missions aimed to perform Earth observation. However, it is important to point out the following discrepancies between the two architectures: ADM-Aeolus is equipped with two surface tension tanks and ten thrusters, while only one bladder tank and eight total thrusters are shown in the scheme.

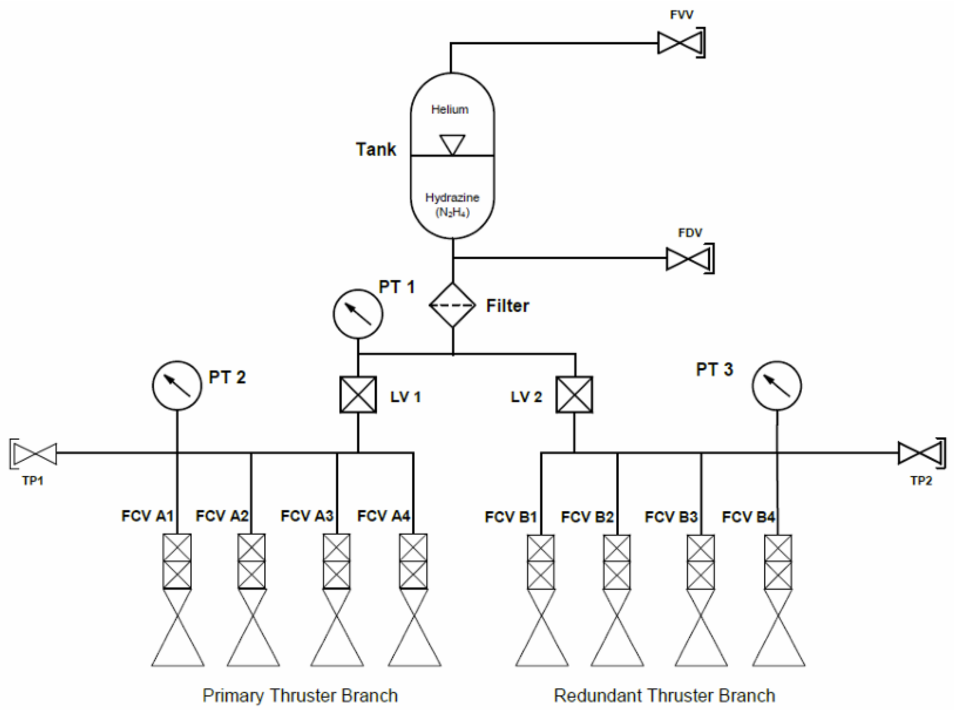
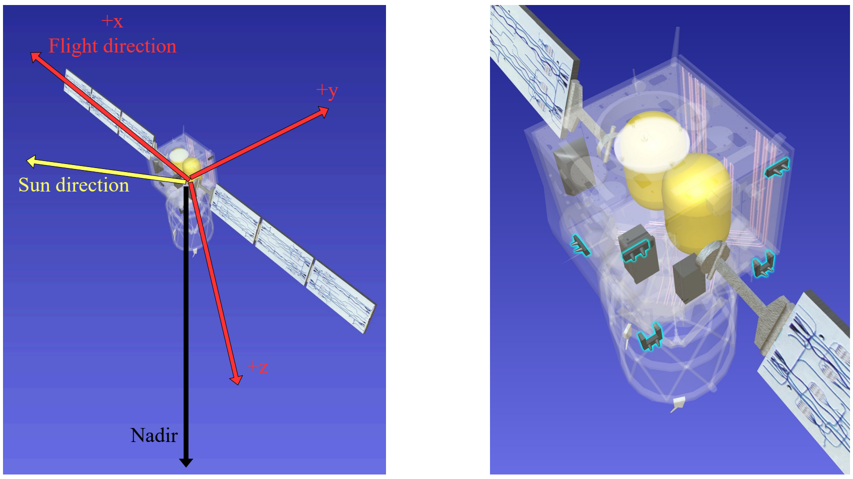


Figure 2.2: Propulsion subsystem schematics.

A description of each component seen in the scheme is reported below.

- **Tanks**: the tanks used in the mission, shown in Figure 2.4a, are the model OST 31-1 from ArianeGroup [19], built in titanium (STA 6Al-4V Ti), which is very resistant to corrosion and thus compatible with hydrazine. The tanks are integrated with a helium blowdown pressurizing system and have 177 L volume capacity each, thus storing 132 L of liquid propellant and 45 L of pressurizing gas. As mentioned before, the tanks are surface tension tanks; therefore, they do not work through a membrane that separates the pressurising gas and the propellant liquid, but with a propellant management device (PMD) that uses the surface tension of the liquid to always keep the propellant towards the exit, preventing the gas from being sucked in place of the fuel. The tanks are placed symmetrically, near the centre of gravity, in order to help balance the satellite during its operational life, even during the effective change of mass as the fuel level decreases. Conversely, placing the tanks far from the centre of gravity would lead to a difficult management of the inertia configuration, since variations in mass away from the centre of gravity would be amplified further by the distance. It is noteworthy to say that this configuration is not the one that reduces the most the load losses in the feeding lines, in fact, it could have been better to place them along the y direction instead of the x direction (Figure 2.3a). On the other hand, such disposition would have reduced the stability on the x axis, affecting the pointing requirement.

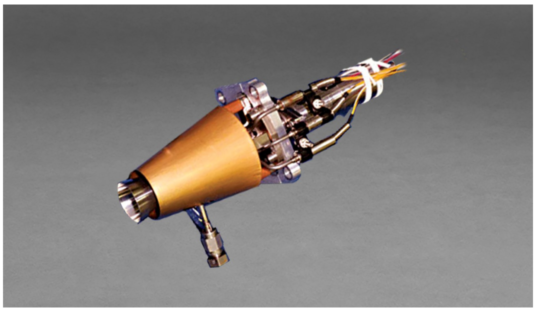
- **Thrusters**: the satellite is equipped with Monarc-5 engines [20][21], shown in Figure 2.4b, that are capable of providing a constant thrust of 5 N with a specific impulse of 230 s. As mentioned before, the thrusters are ten, of which four are used for nominal operations and six are for redundancy. The engine has a limited weight (0.49 kg), which does not significantly impact the overall mass of the spacecraft, keeping the inertia configuration almost unaffected. As depicted in Figure 2.3b, four couples of thrusters, of which one is nominal and the other one redundant, are positioned on the face oriented opposite to the flight direction. Additionally, a fifth redundant couple can be found on the face towards the Sun direction. These thrusters produce a control torque around z axis, thus their presence could be meaningful for RW desaturation. The four nominal couples of thrusters are positioned at the vertices of the surface inclined of 15° [20]. This provides sufficiently high torque, but also protects sophisticated components and solar arrays from the hydrazine flow.
- **FVV and FDV**: the FVV and FDV are valves used to load or drain the pressurizing gas and the propellant from the tank.
- **Filter**: the filter has the functionality of removing solid particles from the hydrazine propellant.
- **PT 1,2,3**: pressure sensors.
- **LV 1,2**: valves that separate the tanks from the two branches of the primary and redundant thrusters.
- **TP 1,2**: are access points to the power grid that allow ground testing operations.



(a) Satellite body axes. (b) Thruster positioning. [22]

Figure 2.3

(a) OST 31-1 tank.



(b) Monarc-5 thrusters.

Figure 2.4: Main components of ADM-Aeolus propulsion subsystem.

2.2.2.1 Propellant selection

Monopropellants offer several advantages, including restart capability and throttleability, enabling precise control during manoeuvres. Hydrazine, in particular, is widely used due to its chemical stability and relatively high performance, with a specific impulse ranging from 200 s to 300 s. In addition, its moderate cost and great versatility facilitate integration into propulsion subsystems, making it an efficient choice for spacecraft architecture, having the advantage of a single-feed line.
Despite the advantages mentioned, hydrazine also presents some environmental issues; in fact, because of its toxicity, alternatives and more environmentally-friendly solutions are being sought. Moreover, the efficiency of the resulting propulsion subsystem is lowered because of the presence of a cathalytic bed, which is needed for the decomposition of the propellant.
Considering the benefits cited above and the operations to be undertaken, hydrazine (N_2H_4) appears to be an optimal choice from the point of view of this mission.

2.2.3 Propulsion subsystem and phases

In the present section, the function and utilization of the propulsion subsystem throughout the different phases of the space mission is analyzed. The propulsion subsystem of ADM-Aeolus described above is **secondary**, since the satellite remains in the nominal orbit throughout its operational life and does not need to perform high thrust manoeuvres. Still, the subsystem is capable of performing station-keeping manoeuvres and generally addressing flight dynamics control. In more detail:

- **LEOP**: despite the expected injection error, the final result was far better, thus no injection errors were left to be corrected through thrusters activation.
- **Commissioning phase**: In this phase the RW are used to support Instrument Calibration (IC), consequently the propulsion unit is exploited to perform RW desaturation. Given this, the importance of having a hydrazine monopropellant as a propulsion system becomes evident, since it has multiple reignition capacity and throttleability, allowing these manoeuvres to be accurate.
- **Operational phase**: Apart from periodical IC manoeuvres, during nominal operations the propulsion system is mainly accounting for station keeping. The latter is highly affected by air drag, thus the thrusters positioning on the face counter-looking the flight direction is justified.
- **End of life**: Finally, although not planned, the assisted deorbiting phase was performed with the remaining propellant. During the phase, the thrusters provided seven impulses aimed at braking the satellite, consequently along the x axis. In order to perform a decelerating manoeuvre, it was necessary to reorient the spacecraft with a rotation of 180° around the y axis, while keeping the Sun arrays Sun-pointing.

Throughout the mission duration, the propulsion subsystem is suitable for all the requested Δv, therefore there is no need to adopt a primary propulsion.

2.2.4 Propulsion subsystem analysis and sizing

In this section, the sizing of the propulsion subsystem is performed. Due to the lack of information regarding the actual mission Δv and since the Δv computed in the previous section is subject to uncertainties, the embarked propellant mass for a single tank, available from literature [11], has been used as a substitute input. Additionally, the data reported in Table 2.2 have been used.

Data			
propellant: hydrazine (N_2H_4)		*tank shape: cylindrical-spherical*	
pressurizing gas: helium (He)		*tank material: Ti-6Al-4V*	
$M_{prop,real}$	133 kg	σ	950 MPa
$P_{\text{gas,i}}$	2.4 MPa	h_{cyl}	0.276 m
ρ_{prop}	1010 kg m^{-3}	ρ_{Ti}	4430 kg m^{-3}
R_{gas}	2077.3 J kg^{-1} K^{-1}	N_{Tanks}	2
$P_{inlet,min}$	0.55 MPa	$N_{Thursters}$	10
ΔP_{feed}	0.035 MPa	T_{tank}	293 K

Table 2.2: Data used for propulsion subsystem sizing.

Starting from the propellant mass, it is possible to compute the corresponding volume, to which a 10% margin is then added to account for unusable volume (MAR-CP-010 [12]):

$$V_{\text{prop}} = \frac{M_{\text{prop,real}}}{\rho_{\text{prop}}} = 0.1317\,\text{m}^3, \quad V_{\text{prop,real}} = V_{\text{prop}} \cdot 1.1 = 0.1449\,\text{m}^3 \tag{2.2.1}$$

Once the propellant volume has been computed, the focus can shift to the sizing of the pressurant. To do so, it is necessary to determine the initial pressure of the gas inside the tank, which, according to literature [19], is equal to 24 bar. The final pressure is then calculated by considering the minimum inlet pressure required by the thrusters [21] and the pressure losses, whose value is assumed due to the lack of data:

$$P_{\text{gas},f} = P_{\text{inlet,min}} + \Delta P_{\text{feed}} = 0.585\,\text{MPa} \tag{2.2.2}$$

At this point, the blowdown ratio can be calculated and then used in the computation of the initial gas volume assuming to have an isothermal transformation:

$$B = \frac{P_{\text{gas},i}}{P_{\text{gas},f}} = 4.1026\,, \quad V_{\text{gas},i} = \frac{V_{\text{prop,real}}}{B-1} = 0.0467\,\text{m}^3 \tag{2.2.3}$$

Helium is used as the pressurant and can be considered an ideal gas. Therefore, its mass is calculated using the ideal gas law, assuming that the pressure of the tank is equal to $P_{\text{gas},i}$ (valid in steady state). Subsequently the mass is increased by a 20% margin (MAR-MAS-090 [12]):

$$M_{\text{gas}} = \frac{P_{\text{gas},i} \cdot 10^6 \cdot V_{\text{gas},i}}{R_{\text{gas}} \cdot T_{\text{tank}}} = 0.18\,\text{kg}, \quad M_{\text{gas,real}} = M_{\text{gas}} \cdot 1.2 = 0.22\,\text{kg} \tag{2.2.4}$$

Given the volumes of propellant and pressurant, it is finally possible to compute the volume of the tank. No margin for the presence of a bladder is added, due to the nature of the considered tank:

$$V_{\text{tank,real}} = V_{\text{prop,real}} + V_{\text{gas},i} = 0.1915\,\text{m}^3, \tag{2.2.5}$$

The tank has a cylindrical-spherical shape, with the height of the cylindrical section being 276 mm, as reported in literature [19]. Using the previously calculated volume and performing numerical calculations, it is possible to determine the radius of the spherical part of the tank.

$$V_{\text{tank,real}} = \pi r_{\text{tank}}^2 \left(\frac{4}{3} r_{\text{tank}} + h_{\text{cyl}} \right) \rightarrow r_{\text{tank}} = 300.2\,\text{mm} \tag{2.2.6}$$

The thicknesses can be easily found by exploiting geometrical relations:

$$t_{\text{tank,sphere}} = \frac{P_{\text{gas},i} \cdot r_{\text{tank}}}{2\sigma} = 0.379\,\text{mm}, \quad t_{\text{tank,cylinder}} = \frac{P_{\text{gas},i} \cdot r_{\text{tank}}}{\sigma} = 0.759\,\text{mm} \tag{2.2.7}$$

The last step is to calculate the mass of a single tank:

$$M_{\text{tank,sphere}} = \rho_{\text{Ti}} \cdot \frac{4\pi}{3} \left[(r_{\text{tank}} + t_{\text{tank,sphere}})^3 - r_{\text{tank}}^3 \right] = 1.91\,\text{kg} \tag{2.2.8}$$

$$M_{\text{tank,cylinder}} = \rho_{\text{Ti}} \cdot \pi \left[(r_{\text{tank}} + t_{\text{tank,cylinder}})^2 - r_{\text{tank}}^2 \right] \cdot h_{\text{cyl}} = 1.75\,\text{kg} \tag{2.2.9}$$

$$M_{\text{tank,tot}} = M_{\text{tank,sphere}} + M_{\text{tank,cylinder}} = 3.66\,\text{kg} \tag{2.2.10}$$

The mass budget is finally determined by incorporating a 10% margin for the harness, as well as the mass of each thruster, which, according to literature, is: $M_{single,thruster} = 0.49\,\text{kg}$.

$$M_{\text{tot}} = 1.1 \cdot (N_{thrusters} \cdot M_{\text{single,thruster}} + N_{tank} \cdot M_{\text{tank,tot}} + N_{tank} \cdot M_{\text{gas,real}}) = 13.92\,\text{kg} \tag{2.2.11}$$

Based on the results obtained and comparing them with the real ones, several observations can be made. First of all, it can be noted that the volume computed for a single tank (191.5 L) is 8.2% greater than the volume of the tanks actually used (177 L). This implies that, in order to meet the total computed required volume (383 L) more than two prescribed tanks would be necessary. This discrepancy can be attributed to the fact that the computations are based on the mass of the propellant on board, which differs from what would have been obtained by using the actual mission Δv, which would have likely been lower. On the other hand, the radius value is very close to the actual value, which, according to the datasheet, is 300 mm.

Another noteworthy aspect is related to the computed mass of the single tank. By comparing the latter with the mass of the actual tank [19], an inconsistency is noted (15 kg vs. 3.66 kg). This difference can be related to the fact that the real sizing of the tank could have been performed considering $P_{\text{gas,i}}$ equal to the burst pressure (48 MPa). According to this workflow and selecting as manufactured thickness the largest between spherical and cylindrical, these results are retrieved:

Parameter	Value
$t_{\text{tank,sphere}}$	0.7585 mm
$t_{\text{tank,cylinder}}$	1.5 mm
$M_{\text{tank,sphere}}$	7.65 kg
$M_{\text{tank,cylinder}}$	3.5 kg
$M_{\text{tank,tot}}$	11.16 kg

Table 2.3: Alternative tank sizing.

As seen in the table Table 2.3 the mass of a single tank is now closer to the real solution. An additional observation is related to the choice of the helium blowdown system for the feeding: its goal is to reduce the complexity and cost of the system, decrease the volume occupied by the tanks (since a tank simultaneously contains both propellant and gas), and increase reliability. Furthermore, helium is chosen over nitrogen because it has a lower density, greater thermal stability, and a lower freezing point.

Chapter 3

Tracking, Telemetry and Telecommand Subsystem

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3.1 Tracking, Telemetry and Telecommand subsystem

3.1.1 Introduction

The ADM-Aeolus mission was conceived within the ESA's Living Planet Programme with the aim of providing **near-real-time data** of the wind profile in the atmosphere. Therefore, it is clear how important the Tracking, Telemetry and Telecommand (TTMTC) subsystem is for the accomplishment of this mission.

3.1.2 TTMTC architecture and justification

In the present section the TTMTC architecture of ADM-Aeolus mission is presented. Data concerning the features of the components in the TTMTC subsystem are very limited, apart from those concerning the antennas; hence, reasonable hypotheses have been made regarding the type of components used, justifying them on the basis of the operations performed by the satellite. The conceptual scheme related to TTMTC is presented in Figure 3.1. Even if not depicted, it is necessary to point out that there are two S-Band antennas, and so two transponders and two diplexers.

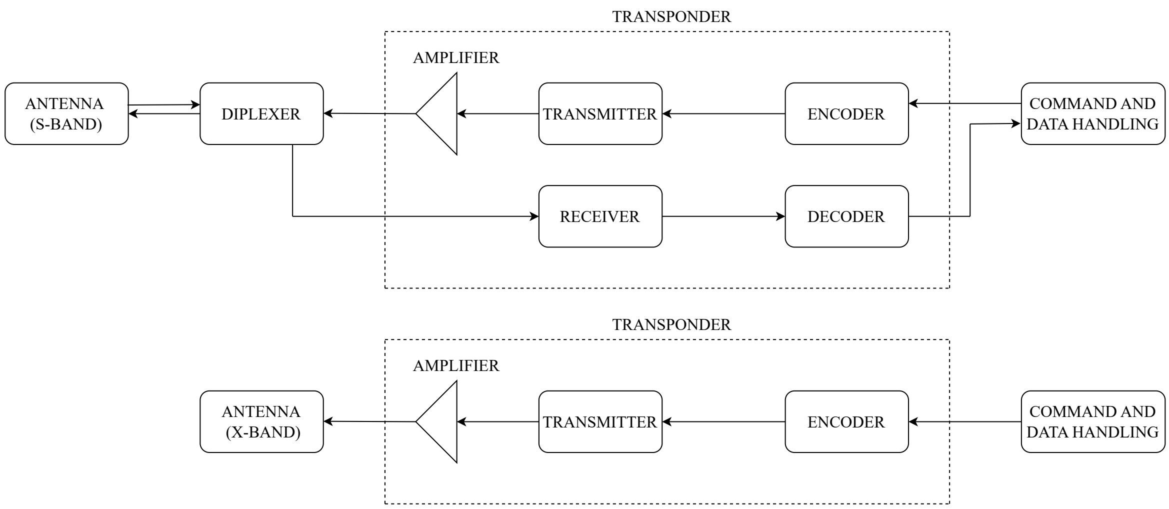

Figure 3.1: Conceptual scheme of the ADM-Aeolus TTMTC subsystem.

Figure 3.2: ADM-Aeolus nadir pointing antennas, with red protective coverings.

- **Quadrifilar Helix Antennas (S-Band):** the spacecraft is equipped with two quadrifilar helix antennas operating in S-Band. The decision of implementing this type of antennas probably

depended on their ability to generate a signal which is naturally circularly polarised. Indeed, having a circular polarization is preferred because the signal remains stable even during continuos attitude changes. The antennas are endowed with opposite circular polarization in order to avoid reciprocal interference; each of them permits a hemispherical coverage thanks to the twisted helical shape [3, Sec. 6.8.2].
The first antenna is nadir pointing and it is meant to communicate with the ground during nominal operations, whereas the other one is pointing along the zenith direction. This configuration, which ensures full coverage, is thought to maintain the link to the ground even in case of failure of the Attitude Determination and Control Subsystem (ADCS)[3]. Since these antennas handle low data rate, they operate in S-Band and are devoted to receive commands and transmit telemetry data to the Flight Operation Segment (FOS) [8];

- **Quadrifilar Helix antenna (X-Band):** scientific data are transmitted through a quadrifilar isoflux helix antenna, which ensures higher compactness with respect to other isoflux solutions [23]. The antenna works at high data rates, which justifies its operation in X-Band and, consequently, the smaller size seen in Figure 3.2. The used antenna ensures a wide coverage; since the satellite must maintain a precise attitude for the correct functioning of its payload, this may be the reason why it was chosen. Additionally, the use of a mechanism to direct the antenna could be prone to technical failure.
- **Diplexer:** since a single S-Band antenna is used to handle both downlink and uplink signals, operating at different frequencies, a diplexer is required to separate the signals without interferences. The diplexer is therefore composed by band-pass filters.
- **Transmitter:** telemetry data are transmitted in S-Band via two transmitters operated in cold redundancy [8]. On the other hand, the scientific data transmitter works in X-Band [11]. In addition, the modulation of the signal is performed in the transmitter. It is reasonable to think that BPSK and QPSK strategies were implemented for their good use of the spectrum and low BER, despite being susceptible to phase disturbances.
- **Receiver:** the spacecraft is equipped with two S-Band receivers operated in hot redundancy [8], meaning that they are both powered simultaneously, so that in case the originally zenith pointing antenna becomes the closest to the nadir direction, continuous communication is ensured.
- **Amplifier:** most of LEO missions have downlink transmission power that varies between 5 W and 10 W. It is fair to suppose that in this case, both for S-Band and X-Band transmission, the power values are in the same range. It is likely that solid-state-amplifiers are implemented, since they are ideal for this power scale, lighter and more robust compared to TWTAs, though less efficient [24].
- **Encoder and Decoder:** The encoder is a device used to protect the data that shall be sent to the ground. Although there is no literature available, it is reasonable to think that the convolutional encoding strategy is used, which has advantages from the point of view of hardware implementation, nevertheless causes a significant increase in bits. The decoder performs the reverse process of the encoder.

3.1.2.1 Ground Station Network

ADM-Aeolus takes advantage of the ESA ground station network. Specifically, the real-time tracking and telemetry data, as well as command data are exchanged with the ground station located in Kiruna (Sweden), which belongs to the Core ESA network, whereas the scientific data and housekeeping data are handled by the station located in Svalbard (Norway), which belongs to ESA's Augmented network. It is worth adding that during the LEOP, due to its criticality, Troll and Svalbard facilities are additionally exploited to supplement real-time telemetry communications, guaranteeing additional coverage and performing the first checks on the health status of the spacecraft [8].

Svalbard ground station is in the polar region. It guarantees an optimal visibility window (up to 15 minutes [25]) and allows for the longest access time to the spacecraft throughout its orbital period, as the satellite follows a Sun-synchronous orbit, which is near-polar. On the other hand, Kiruna station is positioned at lower latitude which results in a shorter visibility window (around 10 minutes [8]) and therefore in lower access time to the spacecraft and in up to five blind orbits per day [7]. In addition, the location of the two stations turns out to be strategic because, given the morphological nature of the two locations, the masking angle is very small, and being very dry places, atmospheric losses are minimized.
Kiruna hosts one 15 m and one 13 m dish antenna. Both operate in S-Band for uplink and downlink and X-Band for downlink. Svalbard Satellite Station (SvalSat) consists of roughly 160 antennas [26], with dish dimensions ranging from 9 m to 13 m, which operate in C-band, L-band, S-Band, X-Band and K-bands [27]. Both facilities are consistent with ADM-Aeolus data transmission.
Finally, it is worth mentioning that, after two years from launch, an additional station in Antarctica was added in order to better comply with the near-real-time data supply requirement [28].

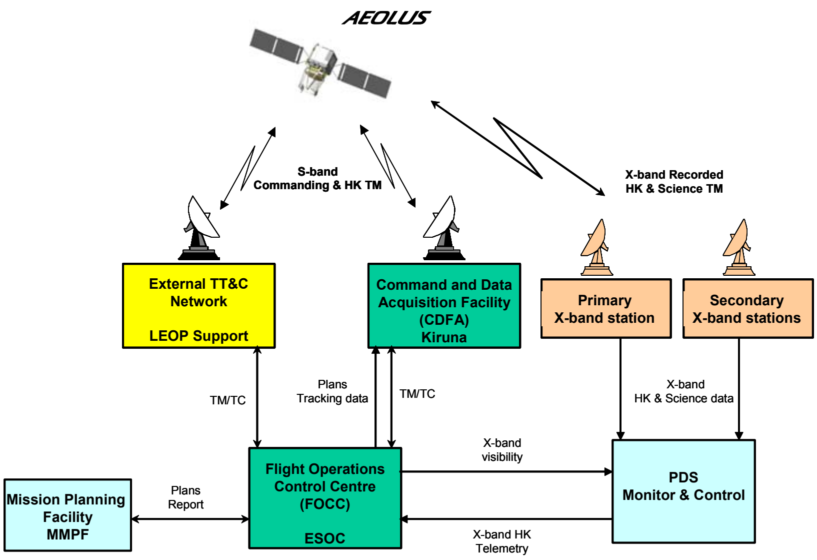

Figure 3.3: Ground station network architecture.

3.1.2.2 Telecommand Strategy and Autonomy

ADM-Aeolus mission aims to continuously send scientific data in quasi-real-time during its operational life. Therefore, having a satellite with high level of autonomy is beneficial, since the Failure Detection, Isolation and Recovery (FDIR) will be able to handle a wide range of critical situations without needing to receive commands. This helps reducing the time without transmitting scientific information during possible issues. The system requirement OPS-27 [3, Sec. 5.2], on 5-day autonomy without ground intervention is therefore justified. In addition, this choice is advantageous to reduce the workload at Kiruna. The alternative to this design choice would have been to employ operators on ground to feed the satellite with the actions to accomplish during criticalities, a much less preferable strategy. Indeed, it is noteworthy to say that lower money expenditure in qualified personnel contributed to the choice of this level of autonomy.

3.1.2.3 Band Selection and Data Store Volume

As stated above, ADM-Aeolus adopts two bands for communications. The S-Band is used for the transmission and reception of telemetry and telecommand. It operates at 2030 MHz in uplink receiving

2 kbps, while the transmission of 8 kbps is carried out at 2205 MHz. Finally the scientific data from ALADIN is sent to the ground station exploiting the X-Band at 8040 MHz with an effective transmission data rate of 5 Mbps [7]. The S-Band is commonly used for low data rate satellite communications, and the Kiruna ground station is equipped with suitable antennas. On the other hand, the X-Band is selected both for the higher data rate with respect to the S-band, and the increased stability towards atmospheric disturbances with respect to higher frequencies [29]. The volume of data exchanged in each band during the effective link time is summarized in Table 3.1. The computation refers to a case where the satellite starts and ends the link at an elevation of 30°. Additionally, the estimation accounts for the whole volume including the encoding, thus it is a clear overestimation of real values which are around a total of 47 MB effective exchanged data [25].

Quantity	Uplink	Downlink	
	S-Band	**S-Band**	**X-Band**
Periodicity	5 d	5 d	90 min
Link time [s]	135	135	135
Data Volume [kB]	33.75	135.00	168750

Table 3.1: Data volumes.

As expected, the heavier data volume is related to scientific data, which are communicated at each passage over the ground station in Svalbard, in accordance with the mission requirements.

3.1.3 TTMTC Phases, operations and modes

The following paragraph links each phase of the mission with the operations that the TTMTC carries out during that phase, with major focus on the active antennas and transmitted signals. Furthermore, each phase will contain a description of the possible operative modes, their main features and how the transition between them is managed.

- **Launch and Early Operations (LEOP)**: after separation, the first steps of the mission include the activation of the S-Band antennas, dedicated to TTMTC, to start transmitting health status signals. During LEOP, according to the system requirements [3, Sec. 5.1.1], the spacecraft shall be able to survive five days without direct communication with ground. In this phase, it is reasonable to think about the existence of a mode in which efforts are not mainly concentrated on the TTMTC, since its importance is overweighted by the need of a proper detumbling and slew maneouvre. Finally, the antenna dedicated to scientific data transmission is off during this phase.
- **Commissioning phase**: in this phase, after verifying the basic functions and the health status of the satellite, ALADIN is turned on and its calibration begins. Concurrently, the X-Band antenna, dedicated to scientific data, is turned on to start the data downlink and to conclude the ALADIN calibration. The S-Band antenna facing the nadir direction continues to communicate TTMTC data, although both S-Band antennas still operate in hot redundancy, as previously mentioned. It is reasonable to assume that, during this phase, a specific mode is implemented in which the TTMTC is in a nominal state, but with the priority of testing the link stability and the functionalities of the main components.
- **Operational phase**: in the Operational phase all three antennas are active. Communications occur in a nominal way and the amount of data exchanged is maximum compared to the other phases. In this stage the Nominal Operation Mode is implemented [3]. It is noteworthy to say that in the case of failure occurrence the satellite shall enter is safe mode. In this case, as mentioned in [3], the TTMTC shall be able to guarantee a two-way communication link with the ground station for at least housekeeping telemetry data and commanding.

- **End of life**: in this phase the TTMTC subsystem supports the end-of-life activities that prepare the satellite for de-orbiting. In particular, commands related to the spacecraft re-entry plan are received and subsequently information related to the correct execution of the de-orbiting procedure is sent. At this stage, scientific data is no longer sampled and the X-Band antenna is used for mass memory dumps, since the satellite is about to burn in the atmosphere [30]. It is reasonable to think that the satellite enters a mode that enhances the effort given to the TTMTC subsystem, to permit the transmission of a large quantity of stored data.

3.2 Link budget estimation

In the present section, the performances of the ADM-Aeolus TTMTC subsystem are evaluated. The main assumptions adopted for the sizing are reported in Table 3.2.

Quantity	**Uplink**	**Downlink**	
	S-Band	**S-Band**	**X-Band**
Transmitted power [W]	100	5	5
Modulation	BPSK	BPSK	QPSK
Encoding	Convolutional	Convolutional	Convolutional
Roll off factor (α)	0.5	0.5	0.5

Table 3.2: Adopted parameters.

The size of the ground antennas is provided, while the dimensions of those on-board the satellite are not available and are therefore estimated on the basis of literature values. From [23] it is understood that the antennas must be contained in a cylindrical protective case of 26 cm height and 10 cm diameter. The antenna models that are deemed appropriate for the mission are reported in [31] for S-Band transmission and in [32] for X-Band transmission. Their sizes are summarised in Table 3.3 and Table 3.4, together with their noise temperature.

Quantity	**AEOLUS**	
	S-Band	**X-Band**
Diameter [mm]	30	9.5
Length [mm]	210	40
Noise temperature [K]	290	290

Table 3.3: Antennas on board ADM-Aeolus.

Quantity	**Ground stations**	
	Kiruna	**Svalbard**
Diameter [m]	13	9
Noise temperature [K]	150	150

Table 3.4: Ground station antennas.

The estimation of the gain and Half-power Beamwidth (HPBW) of the helix antennas is performed using the following equations, taken from [24, Table 13-14]:

$$G_{helix} = 10.3 + 10log\left(\frac{C^2 L}{\lambda^3}\right) \quad \text{where} \quad C = \pi\, D \tag{3.2.1}$$

$$\theta_{helix} = \frac{52}{\sqrt{C^2 L/\lambda^3}} \tag{3.2.2}$$

Although the geometrical parameters of the S-Band helix antenna do not fall within the range of validity of the formulas, the gain and HPBW obtained are consistent with the values found in literature.

The satellite is equipped with low gain antennas and does not change attitude to point towards the ground station, while the ground antennas provide precise tracking. The spacecraft's antenna misalignment is therefore used for the estimation of pointing losses both in the uplink and downlink case.

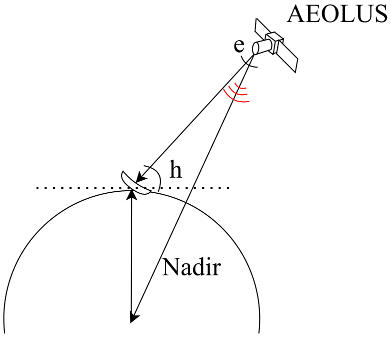

Figure 3.4: Misalignment angle.

The misalignment angle e, shown in Figure 3.4, can be computed through trigonometry and then used to compute the pointing losses. In order to estimate the link budget in a worst case scenario, an elevation angle of $h = 30°$ and a transmitted power from the satellite of $P_{tx} = 5$ W have been used. Therefore, according to the TTMTC subsystem architecture described in section 3.1.2, the S-Band transponder ouputs 10 W, while the X-Band one outputs 5 W. Assuming that the amplifiers mounted on ADM-Aeolus are solid-state models, the overall input power for the S-Band transmission is 50 W, while that for the X-Band transmission is 30 W [24, Fig. 13-15]. Hence, the power requested for the downlink is 80 W, which is a feasible value with respect to the overall power available onboard. Considering convolutional encoding to minimize the Bit Error Ratio (BER), the required E_b/N_0 is recovered [24, Fig. 13-9]. Then, a margin of 3 dB is added according to COM-3 [3, Sec. 6.8.1].

The results of the estimation are shown in Table 3.5. It is worth noting that in all three cases, the values of energy bit over noise spectral density and signal to noise ratio are far above the minimum required values. However, the results might change dramatically for different choices of the dimensions of the antennas on board the satellite. Therefore, in order to provide more precise results, the actual values should be known. Also, as expected, the gain of the quadrifilar helix antennas is not high (it is less than 10 dB in all three cases), but their HPBW is wide, thus providing a good amount of transmitted power for high misalignment angles.

Quantity	**Uplink**	**Downlink**	
	S-Band	**S-Band**	**X-Band**
Transmitted power (P_{tx}) [dB]	20.0	7.0	7.0
TX antenna gain (G_{tx}) [dB]	46.2	9.0	8.7
RX antenna gain (G_{rx}) [dB]	7.9	47.0	55.0
TX antenna HPBW [deg]	0.8	60.4	62.7
RX antenna HPBW [deg]	68.3	0.7	0.3
Cable losses (L_{cable}) [dB]	-2.0	-2.0	-2.0
Pointing losses (L_{point}) [dB]	-7.9	-10.2	-9.4
Atmospheric losses (L_{atm}) [dB]	-3.0	-3.0	-3.0
Free space losses (L_{space}) [dB]	-154.2	-154.9	-166.1
Bandwidth (B) [kHz]	3	12	15000
Net data rate (R_{net}) [kbps]	2	8	5000
Gross data rate (R_{gross}) [kbps]	2	8	10000
Minimum energy bit over noise spectral density (E_b/N_0) [dB]	7.5	8.0	8.0
Minimum signal to noise radio (SNR) [dB]	5.7	6.2	6.2
E_b/N_0 [dB]	78.1	60.8	27.0
SNR [dB]	76.3	59.0	25.3

Table 3.5: Link budget estimation.

Chapter 4

Attitude and Orbit Control Subsystem

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4.1 Attitude and Orbit Control Subsystem overview

4.1.1 Introduction

The ALADIN instrument on-board ADM-Aeolus exploits the Doppler effect to measure the wind velocity along its Line of Sight (LOS). To meet the observational requirements, it is mandatory to have fine knowledge and control of the attitude of the satellite. This constitutes a mission driver and, as will be seen, is an aspect of paramount importance in the design.
As stated in the system requirement AOS-1 [3, Sec. 6.6.1], the satellite shall provide attitude control along three orthogonal axes during all mission phases in order to ensure proper scientific data collection, correctly point the solar panels towards the Sun, and avoid Sun illumination that could damage susceptible components.

4.1.2 AOCS architecture and rationale

In the present section the Attitude and Orbit Control Subsystem (AOCS) architecture is presented. The sensors and actuators on-board the satellite will be described in terms of accuracy, redundancy and position in the spacecraft.

4.1.2.1 Sensors

- **Autonomous Star Tracker (AST)**: The satellite is equipped with two HE-5AS AST, produced by the Danish company Terma [33], which are operated in cold redundancy. They are based on CCD technology and are capable of a measurement accuracy of 13 µrad [7, Sec. 4.4.2]. The ASTs are mounted directly on the ALADIN structure in order to minimize misalignment errors [7, Sec. 4.3.1]. Additionaly, they are located on the side of the satellite that is not exposed to Sun illumination during nominal operations and are pointed towards the zenith direction. This configuration allows the sensors to be protected from sunlight and Earth albedo by the satellite's body, both safeguarding the health of the CCD and ensuring the highest quality of measurement by pointing far from disturbing light sources.
- **Inertial Measurement Unit (IMU)**: ADM-Aeolus features an ASTRIX-200 IMU [34], manufactured by Airbus, which is composed of four Fiber-Optic Gyroscopes (FOGs) arranged in a pyramid configuration. Each of them measures the angular rate along its normal direction. The results are then combined to reconstruct the full angular velocity vector. Indeed, three FOGs along independent directions are sufficient for the correct functioning of the instrument, yet a fourth one is present to ensure internal redundancy and single-point failure tolerance. The instrument is made up of two parts: a head, which contains the actual FOGs and does not produce heat, and an electronic box, which must be cooled down through irradiation. Such configuration has the advantage of allowing the positioning of the measuring head near ALADIN to ensure the accurate measurement of angular rates as close as possible to the payload without the reduction of the performance of the sensor due to the heat produced by its electronics.
- **Rate Measurement Unit (RMU)**: The satellite is also endowed with three QRS-11 MEMS gyroscopes [7, Table 4.4] that are able to provide a coarse estimation of angular rates along three axis. They are lightweight and low power demanding, thus they are used during safe mode instead of the IMU which, on the other hand, requires more energy to function, that might not be available in critical conditions. No redundancy is applied to these components, but they have high reliability, since they do not contain moving parts, and are not critically important for the health of the overall system.
- **Coarse Earth Sun Sensor (CESS)**: Eight CESSs are installed on ADM-Aeolus. Each sensor is made up of a visible and an infrared detector, each featuring three thermistors. Internal redundancy in each detector is achieved by means of a triple voting system that allows to isolate

a single failed thermistor [35]. The sensors are distributed on the spacecraft (three of them are visible on ALADIN structure) so as to guarantee full sky coverage. This allows the satellite to achieve a Sun pointing attitude from every safe mode entry condition. Moreover, they are passive sensors and therefore require very low power to operate, making them optimal for emergency situations.

- **Magnetometer (MTM)**: The spacecraft features two model MAG28 triaxial magnetometers, manufactured by Lusospace [36], operated in cold redundancy [4]. They employ the Anisotropic Magnetoresitance (AMR) technology, which is chosen instead of the fluxgate technology due to its lower cost, production repeatability and easier integration into circuits. Also, the higher accuracy of the latter technology is not necessary since the magnetometer is needed during initial acquisition mode and safe mode for a rough attitude estimate. For the same reason, the sensors are body-mounted and do not use a boom.
- **GPS receiver**: Two identical GPS receivers are installed on the satellite and operated in cold redundancy utilizing the civilian C/A code on the L1 frequency band [7, Table 4.4]. Each unit integrates a RF ASIC and a microprocessor. Such configuration allows an accuracy of under 10 m during nominal mode [7, Sec. 4.4.2]. The receivers provide real-time position, velocity and timing data, allowing orbit determination and synchronization of measurements from other AOCS sensors. This information is also provided to the on-board orbit propagator, which is necessary for the satellite operation schedule, based on its argument of latitude and number of revolutions [8].

It is worth mentioning that, in order to improve measurement accuracy and meet the requirements, the laser ground return detected by the payload is also exploited to complement the information of AOCS sensors [7, Sec. 4.4.2].

4.1.2.2 Actuators

- **Reaction Wheel Unit (RWU)**: The spacecraft is equipped with a set of reaction wheels, which are invaluable for fine pointing control. Full three-axis attitude control requires at least three wheels, indeed, in Aeolus there are four wheels, both for redundancy and to provide greater torque and momentum storage capability. The RWU has an angular momentum storage of 40 N m s and a maximum torque of 0.2 N m [7, Table 4.4]. The wheels are mounted on isolation suspensions to reduce micro-vibrations [37]. In addition, the RWU is located close to the satellite's centre of mass to further mitigate the effects of these disturbances [38]. Among the various 4-wheel redundant configurations, the pyramid configuration is adopted [37]. This arrangement is particularly suitable for ADM-Aeolus, since there is a preferred direction, with each wheel contributing to it, which is the LOS of ALADIN.
- **Magnetorquers (MTQ)**: A set of three magnetic torquers producing momentum on orthogonal axes is mounted on ADM-Aeolus. These torquers provide a rod's field strength of 400 A m^2 [7] and guarantee redundancy using a second winding around the rod [38]. During nominal operations, they are used to unload momentum accumulated by reaction wheels. This is a common choice for a satellite in LEO, since the magnetic field is strong enough to guarantee the performances of the torquers, thus avoiding fuel consumption. In addition, they are employed during safe mode as they are simple, low-power, rugged and reliable actuators [39], and additionally fine pointing is not necessary in this mode.
- **Reaction Control System (RCS)**: ADM-Aeolus utilizes a monopropellant hydrazine propulsion system that features ten thrusters operated in cold redundancy. Four thrusters are employed for orbit control, while two additional thrusters, mounted on a perpendicular face, are probably used for attitude control when larger torque needs to be exerted on the satellite [37]. Further details have been provided in the propulsion subsystem report.

4.1.3 AOCS modes and operations

In order to fulfil the mission objectives, the AOCS subsystem must implement different modes, which enable proper operations and guarantee the different pointing requirements, regardless of the mission phase, the environmental conditions, and the state of the satellite. The modes implemented in the ADM-Aeolus AOCS subsystem and their logic relations are shown in Figure 4.1, while the sensors and actuators used during each mode are displayed in Table 4.1. The modes are subsequently thoroughly described in the bullet points.

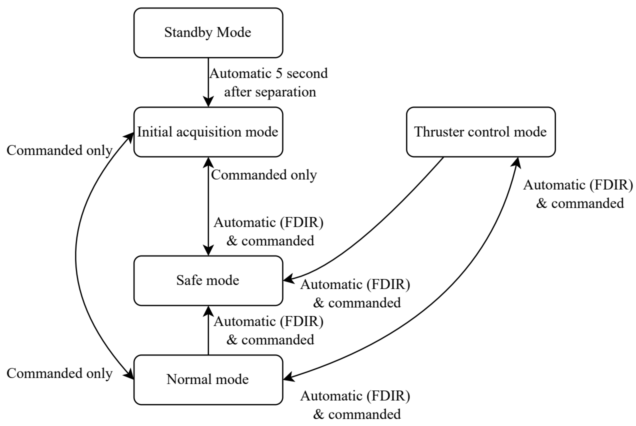


Figure 4.1: AOCS modes logic [35].

Units	SBM	IAM	SM	NM	TCM
CESS		✓	✓		
MTM		✓	✓		
MTQ		✓	✓	✓	
IMU		✓		✓	✓
AST				✓	✓
GPS				✓	✓
RMU			✓		
RCS		✓	✓		✓
RWU				✓	✓

Table 4.1: Sensors and actuators per mode [35].

- **Standby Mode (SBM)**: In this mode, implemented during the pre-launch and launch phase, the AOCS is in a standby state, where sensors and actuators are not active, as shown in Table 4.1. Five seconds after the separation of the spacecraft from the launcher, an automatic transition from SBM to IAM occurs.
- **Initial Acquisition Mode (IAM)**: During this mode the AOCS starts the attitude determination and control process, using the most robust and reliable sensors available: the two magnetometers and the eight CESSs. In this operating state the IMU is also active. Given its operating principle, initial attitude conditions are necessary, they are probably provided by CESSs and MTMs. After separation from the launcher, the spacecraft could experience high angular rates. MTQs provide continuous but slow rate damping, while thrusters are used for faster stabilization when the initial rates exceed the MTQ capability, justifying their use during this mode. The IAM is implemented during the satellite's detumbling and slew manoeuvring and then switched, via telecommand, to the NM. In addition, it is noteworthy that this mode can be triggered not only after the separation, but also in case of need from the safe mode and the normal mode, as depicted in Figure 4.1.
- **Normal Mode (NM)**: It is used by the satellite in nominal condition, when ALADIN is fully functional for the collection of scientific data. In such condition the satellite maintains a flight attitude with ALADIN's LOS tilted 35° with respect to the nadir, in the plane transversal to the velocity direction. This configuration and the inclination of solar panels allow to orient the latter towards the Sun to ensure continuous power generation during atmospheric measurements. It is worth mentioning that throughout the orbit the spacecraft is turned around the yaw axis to compensate for changes in the Earth rotation relative to the flight direction [7, Sec. 4.4.2].
Given the stringent pointing requirements, the AST and the IMU are used as the main sensors. The IMU provides an accurate estimation of angular rates allowing their integration between two

measurements of the AST, which provides precise initial conditions. Regarding control, it is clear that, given the high precision required, reaction wheels are used, so as to guarantee extremely accurate pointing. The use of MTMs during nominal mode is related to the desaturation of the reaction wheels [7, Sec. 4.4.2].

- **Thruster Control Mode (TCM)**: From the literature [13] it can be found that this mode is adopted during orbit maintenance. Therefore the use of thrusters during such mode is evident. In addition, reaction wheels are also active since they are probably used to adjust the satellite attitude before and after performing station keeping.
- **Safe Mode (SM)**: In this mode the CESSs, the magnetometer and the RMU are used as main sensors. The disuse of the AST during this mode is probably related to its high power consumption and slow initialization. Similarly, the IMU also consumes a lot of energy. On the contrary, the use of CESSs and magnetometer is certainly related to their ability to work well in critical conditions, to their low power consumption and their wide FOV. The use of RCS and magnetorquers is linked to the fact that together they are able to cover a very wide range of critical situations remaining reliable.

4.1.4 Pointing Budget

The pointing budget, which is usually built starting from the pointing requirements of the various subsystems, is in this case almost entirely driven by the needs of the payload. No other subsystems have been explicitly identified in the available literature as contributors to the pointing budget. This is perhaps due to the fact that some of them, such as the TTMTC subsystem, which exploits isoflux antennas, do not need a particular orientation to work. The following table shows the pointing budget linked to the ALADIN instrument:

Error Type	**Budget** [µrad]	**Specification** [µrad]
Absolute Pointing, Nominal	301	350
Absolute Pointing, Calibration	299	350
Pointing Stability over 100 orbits	36	40
Pointing Stability over 28 min, Calibration	27	30
Pointing Stability over 10 s	9	10

Table 4.2: ADM-Aeolus LOS pointing budget [7].

The values reported in Table 4.2, including the APE, are very small. It is therefore clear that to satisfy these pointing requirements a 3-axis control is necessary, using actuators for fine control, such as reaction wheels. All of this appears to be consistent with what is used in the satellite during normal mode. No information regarding the AKE can be found, but it is plausible that it is lower than the APE by at least one order of magnitude.

4.2 AOCS reverse sizing

4.2.1 Inertia Matrix

To model the environmental disturbances acting on ADM-Aeolus, and consequently to size the actuators, it is first necessary to obtain the satellite's inertia matrix. For this purpose, a CAD model of the satellite is developed, based on the geometrical dimensions obtained from [22]. Regarding the CAD design, the satellite is modelled considering two different parts: one for the ALADIN and the other for the platform and the solar panels. Each of them is considered in first approximation to have a uniformly distributed mass of 450 kg and 916 kg respectively; the latter value accounting also for

266 kg of propellant. It has to be highlighted that the body axes are approximated with the principal ones, since the extra-diagonal terms are close to zero. The obtained inertia matrix is shown below.

$$I = \begin{bmatrix} 5789.47 & 0 & 0 \\ 0 & 1617.16 & 0 \\ 0 & 0 & 6910.68 \end{bmatrix} \text{kg m}^2$$

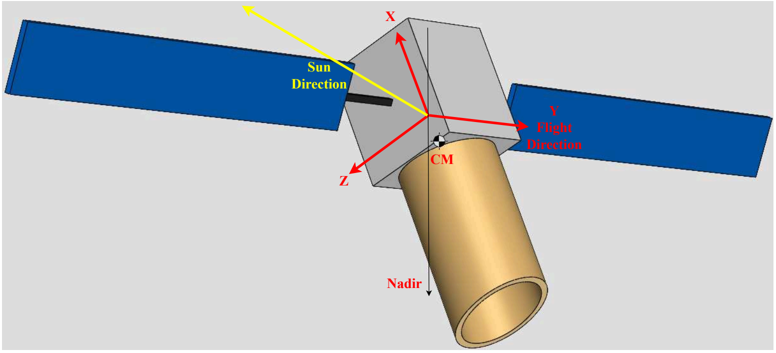


Figure 4.2: Aeolus CAD model.

4.2.2 Environmental disturbances

ADM-Aeolus is in a Sun-synchronous LEO. Therefore, it is reasonable to think that the satellite is subject to strong aerodynamic drag and gravity gradient due to the proximity to the Earth, to a rather strong magnetic field since the orbit is almost polar, and to an almost always present SRP, given the dawn-dusk orbit. In the following sections the disturbances obtained using a Simulink model are presented, with particular focus on the most significant ones. Given the Earth-pointing, it is possible to predict that the air drag and the gravity gradient torques are constant along the orbit, whereas SRP and magnetic field torques are cyclic.

4.2.2.1 Air Drag

Air drag is modelled through Eq (4.2.1)

$$\mathbf{F}_i^{drag} = \begin{cases} -\dfrac{1}{2} C_D \rho(z) v_r^2 \dfrac{\mathbf{v_r}}{\|v_r\|}(\hat{n}_i \cdot \dfrac{\mathbf{v_r}}{\|v_r\|}) A_i & \text{if } \hat{n}_i \cdot \dfrac{\mathbf{v_r}}{\|v_r\|} \geq 0 \\ \mathbf{0} & \text{otherwise} \end{cases} \tag{4.2.1}$$

$$\text{with } \mathbf{v_r} = \mathbf{A}_{B/N}(\mathbf{v}_{S/C}^N - \omega^{Earth} \times \mathbf{r}_{S/C}^N)$$

where $C_D = 2.2$ is considered, as suggested in [13]. In this simplified model, the relative velocity $\mathbf{v_r}$ is computed considering the atmosphere moving alongside the Earth, while the air density is considered to depend only on the altitude. Once the force is computed, it is multiplied by the distance between the centre of mass and the centre of pressure of each surface. The air drag torque is the highest disturbance encountered by the satellite throughout its orbit, since the altitude of the latter is approximately 320 km. Moreover, as seen in Figure 4.3, the disturbance is almost constant throughout the orbital period of the spacecraft. This is due to the fact that the eccentricity of the orbit is close to zero. The non constant behaviour is associated to the fact that the relative wind velocity varies during the orbit.

4.2.2.2 Gravity Gradient Torque

The gravity gradient torque is computed in body axes through Eq (4.2.2)

$$dM = -\underline{r} \times \frac{Gm_t}{|\underline{R} + \underline{r}|^3}(\underline{R} + \underline{r})dm \qquad \begin{cases} M_x = \frac{3Gm_t}{R^3}(I_z - I_y)c_2 c_3 \\ M_y = \frac{3Gm_t}{R^3}(I_x - I_z)c_1 c_3 \\ M_z = \frac{3Gm_t}{R^3}(I_y - I_x)c_1 c_2 \end{cases} \tag{4.2.2}$$

where c_1, c_2, c_3 are the direction cosines of the radial direction in the body frame, while I_x, I_y, I_z are the principal moments of inertia of the spacecraft. Gravity gradient results to be the second most important disturbance affecting the spacecraft's attitude. This is also due to the misalignment of the ALADIN with respect to the nadir direction.

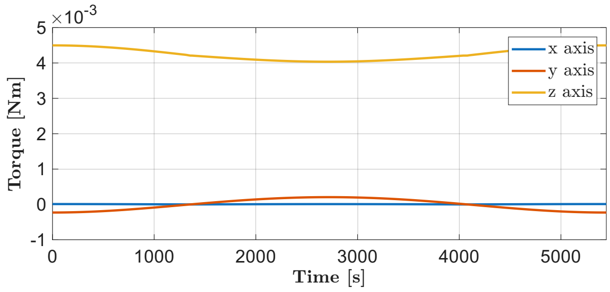


Figure 4.3: Air drag disturbance torque.

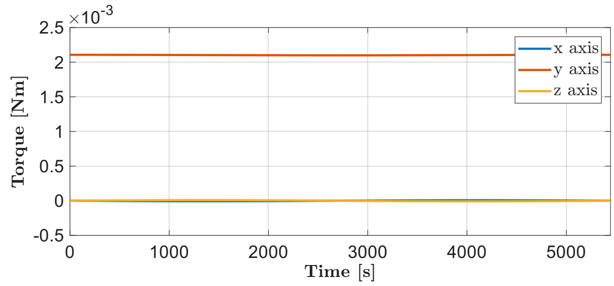


Figure 4.4: Gravity gradient disturbance torque.

4.2.2.3 Solar Radiation Pressure and Magnetic Field Torques

The developed Simulink model takes into account also SRP and magnetic field disturbances. A priori these should be considered, given the low altitude of the orbit for what regards the magnetic field, and the dawn-dusk orbit for what regards SRP. Nevertheless, air drag and gravity gradient overweight their contributions by several orders of magnitude, as can be clearly appreciated in Figure 4.5.

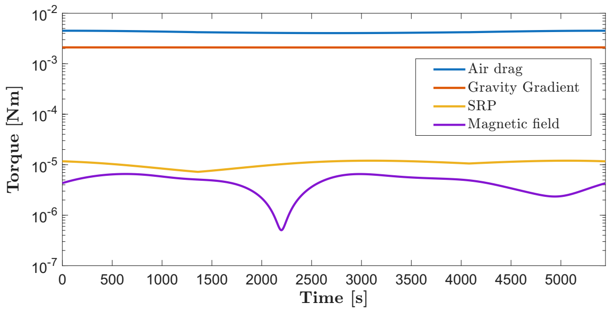


Figure 4.5: Disturbance torques during nominal pointing

4.2.3 Actuators sizing

To cope with the harsh environment to which the satellite is subjected during its mission, it is necessary to use actuators that allow the execution of manoeuvres and, more generally, attitude keeping. In the following paragraphs, the different actuators are sized according to the undertaken manoeuvres. For the computation of the slew manoeuvres an identical angular rate for the acceleration and deceleration phases is considered.

4.2.3.1 Reaction wheels

In the simplified model used in this study, the highest contributions coming from air drag and gravity gradient can be seen to act on different axes. Thus, a sizing of the reaction wheels can be carried out accounting only for the air drag, which has the largest impact. In order to simulate a worst case scenario, the maximum angular momentum stored by a wheel during one complete orbit can be computed by integrating the air drag throughout the orbit, resulting in a $h_{RW} = 23.15\,\mathrm{N\,m\,s}$, in requiring a desaturation every 2.61 h. For the nominal duration of the mission, 3.25 y, a total of 10890 desaturations are needed.
In the reentry phase, the spacecraft is turned 180° multiple times for thruster pointing [14]. Thus, an iterative process, which takes into account the maximum admissible slew rate of $\dot{\theta}_{slew} = 0.5\,°\,\mathrm{s}^{-1}$ [40] and the maximum feasible angular momentum, is carried out to find the correct torque to be applied by the wheels for a correct slew manoeuvre. The results can be appreciated in Figure 4.6.

4.2.3.2 Thrusters

The thrusters are mainly used for orbit maintenance, as discussed in the previous report on the propulsion system. The thrusters can nevertheless be sized according to the slew manoeuvres to check their performance. Their positions and the according distances from the spacecraft's centre of mass are found in [20]. This information can be used to compute the torque to have the minimum time for a slew of 180°. An acceleration and deceleration time $t_{acc} = t_{dec} = 3.7\,\text{s}$ and a coasting time $t_{coast} = 356.3\,\text{s}$ are found, leading to a total slew time $t_{slew} = 363.7\,\text{s}$. The resulting slew manoeuvre is displayed in Figure 4.7. A single slew manoeuvre requires 0.065 kg of propellant, meaning 0.078 kg applying margins.
The wheel's desaturation is carried out at a reduced thrust due to feasibility reasons, in order to have a minimum of 5 s of desaturation process. This would require a propellant mass of 58.7 kg to carry out the desaturations. This resulting value is then increased to account for a 20% margin [3, Sec. 6.6.4], making a total of 70.5 kg of propellant. The final result is suspiciously high, suggesting that much of the desaturation process is carried out by the magnetorquers.

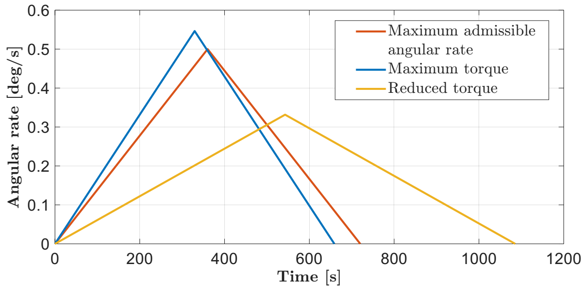


Figure 4.6: Reaction wheels slew.

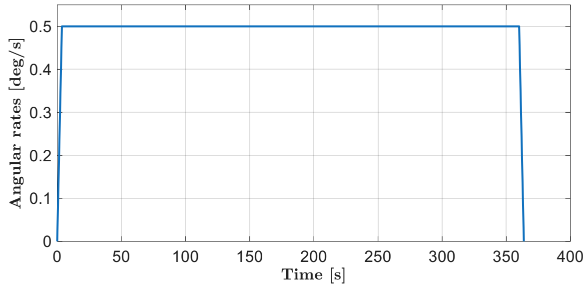


Figure 4.7: Thruster slew.

4.2.3.3 Magnetorquers

Magnetorquers' effectiveness depends on the intensity of the magnetic field encountered by the satellite. An accurate magnetic field model [41] is used to carry out a control simulation in order to compute the maximum dipole required by the actuators to withstand the external disturbances. The result shows a required dipole equal to $D = 297\,\text{A}\,\text{m}^2$, which is easily delivered by the on-board actuators. The use of magnetic dipole for desaturation is modelled using a mean value for external magnetic field and the maximum dipole available, resulting in a minimum desaturation time of 1 min.

4.2.4 Mass, data and power budget

Once the sizing is carried out, a total mass of 382.39 kg is computed using datasheet information [42][43][44]. This value accounts for the complete propulsion system and propellant. Additionally, the estimate of the data budget is performed, exploiting typical values found in literature [24]. Finally, the power budget per mode is retrieved. The results are shown in Table 4.3.

Quantity	**Actuator**			**Sensor**					
	RWU	**RCS**	**MTQ**	**AST**	**IMU**	**RMU**	**CESS**	**MTM**	**GPS**
Mass [kg]	26.8	304.87	23.4	6	12.7	≤ 0.18	0.64	0.6	7.2
Data [kbit]	4.9	6.6	3.3	245.8	16.4	8.2	13.1	1.6	N/A

	SBM	**IAM**	**SM**	**NM**	**TCM**
Power [W]	0	175.3	146.5	185.8	284.5

Table 4.3: Actuators and sensors budgets.

Chapter 5

Thermal Control Subsystem

Contents

Change log	
Issue	*Comments*
1	First release

5.1 Thermal Control Subsystem overview

5.1.1 Introduction

The Thermal Control Subsystem (TCS) is an essential element of ADM-Aeolus, since it counteracts internal and external thermal disturbances. Indeed, during its operational lifetime, the satellite is subjected to external thermal fluxes incoming from the Sun, Earth albedo and Earth infrared emission, as well as internal emission from the payload and on-board electronics. More in depth, the TCS provides the thermal environment required to ensure the integrity and the proper performance of the satellite during LEOP, the nominal phase of the mission, safe mode and initiation of any end of life manoeuver, as stated in system requirement TCS-7 [3, Sec. 6.4.2]. In the following sections, the solutions adopted in the TCS are discussed, verifying that the main requirements are satisfied. Subsequently, a preliminary sizing is carried out.

5.1.2 Temperature ranges for main components

The TCS is designed to maintain each component within the temperature range required for correct operation when active, or survival when in idle state.

S/S	**Component**	**Operating**		**Non-operating**		**Source**
		T_{min} [°C]	T_{max} [°C]	T_{min} [°C]	T_{max} [°C]	
P/L	ALADIN [45]	20	25	-15	50	Exact
PS	Hydrazine tanks [46]	10	50	10	50	Analogous
TTMTC	Antennas [47]	-150	150	-150	150	Analogous
AOCS	AST [48]	-40	70	-40	70	Exact
	IMU [34]	-10	50	-20	60	Exact
	RMU [49]	-40	80	-55	100	Analogous
	CESS [50]	-175	150	-175	150	Analogous
	MTM [36]	-30	60	-40	70	Exact
	MTQ [51]	-35	65	-40	80	Analogous
	RWU [42]	-15	60	-15	60	Analogous
	GPS receivers [44]	-20	60	-20	60	Analogous
EPS	Batteries [24]	0	15	-10	25	SMAD
	Solar arrays [24]	-150	110	-200	130	SMAD
	PCDU [52]	-35	70	-35	70	Analogous
OBDH	Processor [53]	-55	125	-65	150	Analogous
	Memory [54]	-40	105	-55	150	Analogous

Table 5.1: Satellite components temperature ranges.

Table 5.1 shows the temperature ranges required by the main elements on-board ADM-Aeolus. Data is retrieved from the datasheets of components installed on the satellite (Exact) or on analogous missions (Analogous), and from [24] (SMAD). Whenever not present, non-operating temperature ranges are set equal to the operating ranges for a conservative approach. It can be seen that the most stringent requirement is imposed by the payload. This is reasonable since ALADIN exploits a laser, which requires a stable temperature environment both to generate a monochromatic beam and to accurately sense the backscattered signal. For this reason, as will be further detailed, the payload is the only element of the satellite that requires the implementation of active thermal control strategies.

5.1.3 TCS design

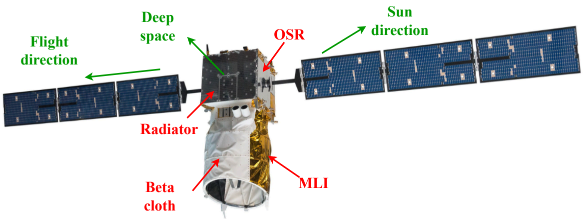
As mentioned in system requirement TCS-9 [3, Sec. 6.4.3], the thermal control of the platform and of the payload should be achieved by passive means and with the aid of electrical heaters where necessary. The passive and active solutions adopted for the TCS architecture and their rationale are now detailed.

5.1.3.1 Passive control

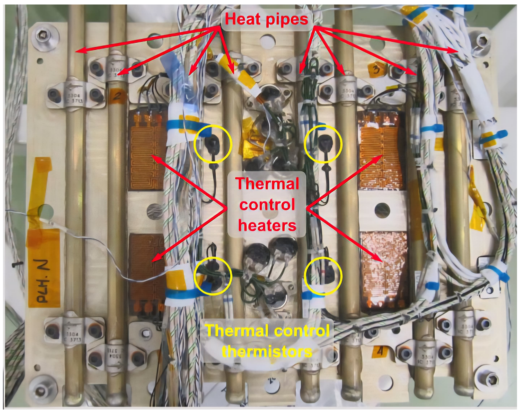
- **Radiator**: Radiators are passive devices used to dissipate heat into space through irradiation. In ADM-Aeolus a large radiator with a surface area of $2\,\mathrm{m}^2$ is used [55]. It is part of the Laser Cooling System (LCS) of ALADIN and ensures that, during its operational life, the lidar does not exceed the thermal boundaries. ALADIN requires very high power (840 W [56]) and, like all laser-based systems, it has limited efficiency [57]. Therefore, a large part of the power is transformed into heat, thus justifying the large dimensions of the radiator. Although not confirmed by the literature, through a visual analysis it can be stated that a stood-off radiator is used. This may be motivated by the greater flexibility in integration compared to a panel mounted radiator, since the main structure is derived from a previous mission, Mars Express [9]. The radiator embeds heat pipes in the main panel, probably composed of aluminium skins with a honeycomb core. This choice allows to reach sufficiently high emissivity levels (> 0.8) and therefore to remove high amounts of power (up to 600 W [55, Tab. IV-1]). The radiator is located on the anti-Sun side of the platform, as shown in Figure 5.1a. In this way, it is always oriented towards deep space, therefore irradiation is maximized. Finally, additional minor radiators are dedicated to Detection Front-end Units (DFUs) [7, Sec. 4.3.5.5].
- **Heat Pipes (HP)**: Heat pipes are channels inside which a fluid in phase transition, usually ammonia, flows. In the ADM-Aeolus mission they contribute to the functioning of the LCS, allowing to transport heat coming from the laser heads of ALADIN towards the radiator, as depicted in Figure 5.1c. Their use is justified by the fact that they work very efficiently, exploiting capillarity, an ideal choice in the absence of gravity, and because they represent a mature and reliable technology. Twenty heat pipes, probably made of aluminium, are exploited in the LCS [7, Sec. 4.3.3.1] and their length is approximately around 1.5 m to 2 m, sufficiently long to transport the dissipated heat generated by the lidar to the radiator. In addition, further heat pipes are located in the DFUs, working in combination with the dedicated radiators [7, Sec. 4.3.5.5].
- **Cold Plate (CP)**: CPs are metal elements, usually made of aluminium or copper, used to absorb and transfer heat through conduction. They are used in the Power Laser Head (PLH) and Reference Laser Head (RLH) of ALADIN. As shown in Figure 5.1c, they are connected to the heat pipes, so as to transfer heat to them.
- **Multi Layer Insulator (MLI)**: ADM-Aeolus uses MLI blankets to insulate the external surfaces from orbital temperature extremes. The coating, which appears gold, is probably made of aluminized Kapton. This material is particularly suitable for satellites operating in LEO, since Kapton has a moderate solar absorptance and a high IR emittance [58]. It is relevant to notice that these blankets are primarily used on the face of the main body exposed to the Sun, as shown in Figure 5.1a, as they reflect solar radiation and minimize heat absorption.
- **Beta cloth**: Beta cloth consists of thin glass fibers woven into a fabric impregnated with Teflon [59]. Although its main scope is to shield ALADIN from atomic oxygen, it partially contributes to thermal control as well, thanks to its low solar absorptance and high emittance [58]. It is reasonable to assume that this coating is used in combination with MLI, which provides better thermal stability but is more susceptible to degradation. For this reason, beta cloth is used only where protection against the space environment is crucial, namely the surfaces in the direction of motion. On the opposite side, less subjected to erosive action, only MLI is present, as shown in Figure 5.1a, since adding beta cloth would increase the weight without a real need for mechanical protection.
- **Optical Solar Reflector (OSR)**: The external areas of the ADM-Aeolus platform are also covered by OSRs, typically made of metallized quartz [60]. As depicted in Figure 5.1a, they are positioned on the lateral surfaces, where electronic components, like the PCDU and the

CDMU, are installed. This choice is consistent with the properties of the OSRs of reflecting solar radiation and emitting excessive internal heat caused by on-board systems. This solution aims to manage thermal control of the electronic components without the need for active cooling.

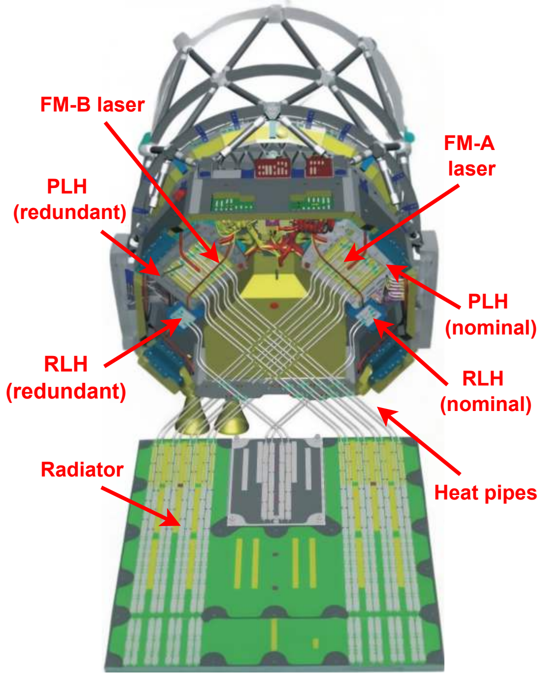
- **Materials**: The main structure is composed of aluminium sandwich panels with an internal honeycomb core [7, Sec. 4.4.1], also made in aluminium, which has good conductivity and mechanical properties; this contributes to the thermal control of the platform. Solar arrays are covered with CMO glass type [61], an antireflective coating, that serves also as thermal protection, even though it is not its main functionality. Whereas, the surfaces of the panels on the anti-Sun direction are probably coated with white paint, which allows to dissipate excessive heat, given its high emittance [62].



(a) ADM-Aeolus external view.



(b) ADM-Aeolus PLH [63].



(c) ALADIN TCS [63].

Figure 5.1: Overview of ADM-Aeolus spacecraft.

5.1.3.2 Active control

- **Heaters**: ADM-Aeolus is equipped with heaters, devices able to generate power through the Joule effect, which are installed on both the laser transmitter and receiver. They work in combination with thermistors through feedback PID control [64] and are required since the payload imposes strict thermal ranges. Six heater lines are installed on the primary mirror [65], where a constant temperature is crucial to minimize thermal deformation and thus measurement bias [64]. Another set is installed on the secondary mirror struts and is used to finely adjust the telescope's focus through thermal expansion [64]. Finally, four sets of heaters are installed on the cold plates to regulate the PLH and RLH temperature, as shown in Figure 5.1b, and work in coordination with the passive heat pipes to ensure 1 K accuracy [7, Sec. 4.3.3].
- **Thermo Electric Coolers (TECs)**: The laser backscattered signal is sensed by CCDs, which operate at a temperature of $-30\,^{\circ}\mathrm{C}$ to limit dark-current noise. To ensure the proper working condition, they feature integrated Peltier effect TECs, whose excess heat is transported to the DFU radiators via heat pipes [7].

5.2 Thermal Control Subsystem reverse sizing

5.2.1 Hot case and Cold case scenarios

In order to correctly carry out the preliminary sizing of the TCS, the worst case scenarios faced throughout the mission shall be identified. To properly identify the scenarios, it is important to note that, since the mission operates in a circular LEO, the only factor affecting the external heat fluxes is whether the satellite is in eclipse or not. In particular, when the satellite is in sunlight, it is exposed to solar, albedo and infrared radiation. Instead, when the spacecraft is in eclipse for up to 20 minutes per orbit [66], only the infrared radiation coming from Earth is present as external heat source.
Regarding the internal heat power, the worst case scenarios can be identified by estimating the heat generated by each subsystem in each mode. The data are obtained considering that the antennas and the payload dissipate part of the total power required, since they involve the transmission of electromagnetic waves, whereas the other subsystems dissipate all the required power. The results are listed in Table 5.2, with the Communication Mode (COM) being the most critical in terms of generated heat, thus associated to the hot case, and with the Safe Mode (SM) associated to the cold case.
The first result can be motivated by the following considerations: ALADIN is the most power-demanding component on-board, thus also the largest heat generator and therefore it must be active for the identification of the hot case; all the antennas are actively transmitting and the AOCS is in normal mode, thus contributing to the generation of heat. The power dissipated internally, accounting for all the components active in this mode, has a value of $Q_{\text{int,hot}} = 1083\,\text{W}$.
The second result can be motivated by the fact that, during SM, the satellite strongly limits non-essential operations. Therefore, ALADIN is switched off, the AOCS has lower energy consumption and only S-band antennas are active. The resulting internal heat power is therefore $Q_{\text{int,cold}} = 347\,\text{W}$.
Combining the information about external and internal heat sources, it is possible to conclude that the hot case occurs when the satellite is in Communication Mode and exposed to sunlight, while the cold case occurs when it is in Safe Mode during eclipse.

S/S	LEOP	Commissioning	Operational				EOL
			SCM	TCM	COM	SM	
P/L [W]	0	672	672	0	672	0	0
PS [W]	0	0	0	40	0	0	40
TTMTC [W]	40	65	0	0	65	40	65
AOCS [W]	147	186	186	285	186	147	285
EPS [W]	120	120	120	120	120	120	120
OBDH [W]	40	40	40	40	40	40	40
$\mathbf{Q}_{\text{int}}$ [W]	347	1083	1018	485	1083	347	550

Table 5.2: Internal heat power generated per mode.

5.2.2 Single-node approach

For the preliminary sizing of the TCS a single-node approach is employed. Since the solar arrays are connected to the main body with small rods and have wide operating temperature range, they are not included in any energy balance, as it is expected that they are able to regulate their temperature passively. In the model, a spherical approximation is used, with an area matching the one of the satellite. The maximum and minimum temperature are identified following the operating temperatures of the components in Table 5.1. The range $-30\,^\circ\text{C}$ to $70\,^\circ\text{C}$ is considered to accommodate the strong approximation of the single-node analysis. Additionally, this range is further reduced to $-15\,^\circ\text{C}$ to $55\,^\circ\text{C}$ by considering a margin of $15\,^\circ\text{C}$ (MAR-TCS-010 [12]). At this point the heat fluxes are computed for both the hot case and the cold case. Regarding the hot case, the contributions of the Sun radiation, the Albedo and the infrared radiation are computed. Once these values are known, the internal dissipated power is added and the temperature of the spacecraft in the hot case is then

retrieved. For a better analysis, several iterations with different values of emissivity and absorptivity are carried out.
The temperature reached by the spacecraft is then compared with the maximum allowed margined value, and since it exceeds it, the presence of a radiator is considered. The areas of the radiators are then retrieved with Eq (5.2.1), where A_{space} is the area of the spacecraft radiating towards the deep space, recovered as the difference between the total surface and the portion of area facing Earth.

$$A_{\text{rad}} = \frac{Q_{\text{sun}} + Q_{\text{albedo}} + Q_{\text{ir}} + Q_{\text{int,hot}} - \sigma \cdot \varepsilon_{\text{sc}} \cdot A_{\text{space}} \left[(T_{\text{sc}}^{\max})^4 - T_{\text{space}}^4 \right]}{\sigma \cdot (\varepsilon_{\text{rad}} - \varepsilon_{\text{sc}}) \left[(T_{\text{sc}}^{\max})^4 - T_{\text{space}}^4 \right]} \tag{5.2.1}$$

The cold case scenario must then be sized. In this situation the only contributions to the power balance are the infrared radiation and the internal heat in SM. It is also important to consider the contribution of the radiator in the emitted power. The temperature for this configuration is then computed. Since it is much lower than the minimum acceptable value, heaters shall be considered in the sizing. The sizing equation to find the power required by the heater is found imposing the minimum margined temperature. A margin of 25% is later applied.

$$Q_{\text{heat}} = \sigma \cdot \varepsilon_{\text{sc}} \cdot (A_{\text{space}} - A_{\text{rad}}) \left[\left(T_{\text{sc}}^{\min}\right)^4 - T_{\text{space}}^4 \right] + \sigma \cdot \varepsilon_{\text{rad}} \cdot A_{\text{rad}} \left[\left(T_{\text{sc}}^{\min}\right)^4 - T_{\text{space}}^4 \right] - Q_{\text{ir}} - Q_{\text{int,cold}} \tag{5.2.2}$$

The data used for the design, along with the output results, are listed in Table 5.3.

$\varepsilon_{\mathbf{sc}}$ [-]	$\alpha_{\mathbf{sc}}$ [-]	$\varepsilon_{\mathbf{rad}}$ [-]	$\mathbf{T}_{\text{node}}^{\text{min}}$ [°C]	$\mathbf{T}_{\text{node}}^{\text{max}}$ [°C]	$\mathbf{Q}_{\text{sun}}$ [W]	$\mathbf{Q}_{\text{albedo}}$ [W]	$\mathbf{Q}_{\text{ir}}$ [W]	$\mathbf{A}_{\text{rad}}$ [m^2]	$\mathbf{Q}_{\text{heat}}$ [W]
0.3	0.35	0.85	-15	55	3139	1581	773	8.84	1748

Table 5.3: Single-node data and results.

The performed iterations yield unacceptable values in terms of radiator's area and required heating power, since they are not suitable for the size and power availability of the satellite. Table 5.3 presents the results corresponding to the minimum radiator area obtained, which is anyway larger than the actual value. The power requested from the heaters is computed considering the same radiator area and it is far higher than the one actually used, as seen in Table 5.5. The values of emissivity and absorptivity shown in Table 5.3 are consistent with values from literature. It is possible to conclude that the single node analysis highlights the need of radiators and heaters, but for refined results a multi-node analysis is needed.

5.2.3 Three-node approach

In this section a multi-node approach based on three nodes is considered. Each node is representative of the actual panels of the spacecraft facing the Sun, the Earth and deep space respectively. Each face emits and receives radiation from the others and all of them radiate externally to deep space. A different value of emissivity is considered for each node, in order to better represent the material of the surfaces of the satellite. The view factors $F_{i \to j}$ result from a simplification based on a triangular-faced prism of infinite length. The modelling equations in the hot case are expressed in Eq (5.2.3), where the internal coating has been approximated as a grey surface, so that $\varepsilon_{\text{int}} = \alpha_{\text{int}}$. In the cold case, the system is the same with $Q_{sun} = Q_{albedo} = 0$ and Q_{int} being the one in SM. All three nodes are constrained to the same temperature range as that of the single node analysis.

$$
\begin{cases}
Q_{\text{int},1} + Q_{\text{sun}} - (A_1 - A_{\text{rad},1})\sigma\varepsilon_1(T_1^4 - T_{\text{space}}^4) - A_{\text{rad},1}\sigma\varepsilon_{\text{rad}}(T_1^4 - T_{\text{space}}^4) + \\
\quad -\varepsilon_{\text{int},1}\sigma A_1(F_{1\to2}T_1^4 + F_{1\to3}T_1^4) + \alpha_{\text{int},1}\sigma(F_{2\to1}A_2T_2^4 + F_{3\to1}A_3T_3^4) + \\
\quad -h_{\text{con}}A_{c_{1\to2}}(T_1 - T_2) - h_{\text{con}}A_{c_{1\to3}}(T_1 - T_3) = 0 \\
Q_{\text{int},2} + Q_{\text{albedo}} + Q_{\text{ir}} - (A_2 - A_{\text{rad},2})\sigma\varepsilon_2(T_2^4 - T_{\text{space}}^4) - A_{\text{rad},2}\sigma\varepsilon_{\text{rad}}(T_2^4 - T_{\text{space}}^4) + \\
\quad -\varepsilon_{\text{int},2}\sigma A_2(F_{2\to1}T_2^4 + F_{2\to3}T_2^4) + \alpha_{\text{int},2}\sigma(F_{1\to2}A_1T_1^4 + F_{3\to2}A_3T_3^4) + \\
\quad -h_{\text{con}}A_{c_{2\to1}}(T_2 - T_1) - h_{\text{con}}A_{c_{2\to3}}(T_2 - T_3) = 0 \\
Q_{\text{int},3} - (A_3 - A_{\text{rad},3})\sigma\varepsilon_3(T_3^4 - T_{\text{space}}^4) - A_{\text{rad},3}\sigma\varepsilon_{\text{rad}}(T_3^4 - T_{\text{space}}^4) + \\
\quad -\varepsilon_{\text{int},3}\sigma A_3(F_{3\to1}T_3^4 + F_{3\to2}T_3^4) + \alpha_{\text{int},3}\sigma(F_{2\to3}A_2T_2^4 + F_{1\to3}A_1T_1^4) + \\
\quad -h_{\text{con}}A_{c_{3\to1}}(T_3 - T_1) - h_{\text{con}}A_{c_{3\to2}}(T_3 - T_2) = 0
\end{cases}
\tag{5.2.3}
$$

The first step of the analysis involves computing temperatures in both scenarios without any heater nor radiator. As expected the spacecraft is not thermally self-sustained without further aid, thus radiators are included in the analysis. The results of the second step contain negative values of the areas. This implies that the radiator needs to generate heat to satisfy the temperature constraint, which defeats its purpose. Therefore radiators on the nodes with negative area are not considered and a new set of maximum temperatures is computed. Finally, the cold case is studied with the corresponding internal and external heat fluxes. The required power from the heaters is obtained including the margin, as done for the single node.

Node	ε [-]	α [-]	$\varepsilon_{\mathbf{rad}}$ [-]	$\mathbf{T}_{\text{node}}^{\text{min}}$ [°C]	$\mathbf{T}_{\text{node}}^{\text{max}}$ [°C]	$\mathbf{A}_{\text{rad}}$ [m^2]	$\mathbf{Q}_{\text{heat}}$ [W]
1	0.3	0.35	0.85	-15	45.2	3.3	960
2	0.56	0.37	0.85	-15	39.1	0	30
3	0.3	0.35	0.85	-15	-4.1	0	622

Table 5.4: Three-node data and results.

As shown in the Table 5.4, the results are well below the maximum admissible temperatures. The presence of a single radiator, cooling one node, has a broader effect on the whole system of nodes, yielding more reasonable results. The required heating power is still high with respect to what the spacecraft is able to produce. For both approaches, neglecting time transients and requiring large areas, rather than single components, to be kept at constrained temperatures, produce solutions that are by far different from what is actually found on-board.

5.2.4 Mass, data and power budget

The budgets related to the TCS are shown in Table 5.5. The mass budget is found to be equal to 4% of the total dry mass, consistent with what is stated in [24]. The data budget, expressed per single sample, is mainly associated to thermistors, and it is taken from [24]. Finally, the power budget, mainly associated to the heaters, is estimated considering the presence of twenty heaters with a required power of 10 W each.

Quantity	**Radiator**	**HP**	**CP**	**MLI**	**Beta cloth**	**OSR**	**Heaters**	**TEC**	**Total**
Mass [kg]	7	6	10	13	3	1.4	negl.	negl.	40.4

Quantity	**TCS**
Data [kbit]	25
Power [W]	200

Table 5.5: TCS budgets.

Chapter 6

Electric Power Subsystem

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6.1 Electric Power Subsystem analysis

6.1.1 Introduction

The Electric Power Subsystem (EPS) is intended to produce, control, store and distribute electrical power within the satellite, as mentioned in system requirement EPS-4 [3, Sec. 6.5.2]. The power is generated by solar panels and stored in a battery, which helps provide the required power during all phases and modes of the mission, when needed. More in depth, the subsystem must be able to provide the majority of the power generated to ALADIN, for its correct operation, as well as allowing all other subsystems to function properly.

6.1.2 EPS architecture and rationale

The probable architecture of the ADM-Aeolus's EPS is presented in Figure 6.1, based on the data found in the literature. The main components, along with their functions and specific features, are also discussed hereafter.

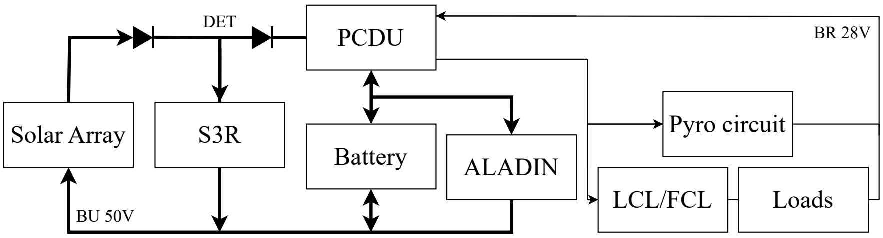

Figure 6.1: EPS architecture scheme.

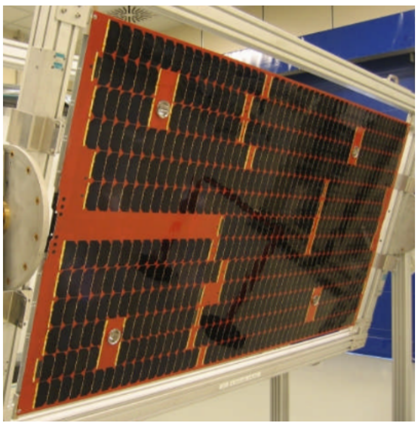

Figure 6.2: Solar Array cells.

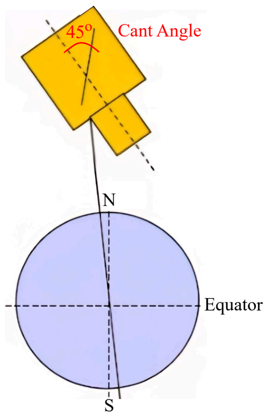

Figure 6.3: Cant angle.

- **Solar Array (SA)**: ADM-Aeolus is on a Sun-synchronous LEO, at one AU from the Sun. Solar panels are therefore a straightforward solution for generating electrical power on-board, and constitute the primary electric power source. Since the satellite orbits the Earth, the use of a Radioisotope Thermoelectric Generator (RTG) is discarded a priori for safety reasons.
  The SAs provide the bus with a maximum voltage of 52.5 V and enable the recharge of the battery. They are capable of producing a peak power of 2.4 kW, which meets the 2238 W requirement at EOL during Summer Solstice [67], leaving a surplus of approximately 1 kW, since the average power consumption is 1.4 kW [7].
  The solar wings are deployed and reach a cant angle of 45° through a Solar Array Rotation Mechanism (SARM) [68]. This angle, depicted in Figure 6.3, is defined as the angle between the

spacecraft's main body direction and the panel's surface and its value is intended to maximize the exposure of SAs to sunlight. Although the satellite must maintain a constant attitude, the incidence of solar radiation on the panels varies throughout the orbit. Consequently, a 45° cant angle is chosen to ensure the most stable power delivery possible. In addition, the use of fixed solar panels helps minimize mechanical complexity. The SAs embed Triple-Junction Gallium Arsenide (GaAs) solar cells, which offer a high conversion efficiency of approximately 27% [67]. To maximize the spectral response of multi-junction solar cells, CMO glass type with antireflective coating is employed.
Each panel is composed by 19 strings in parallel, each of 30 cells, as seen in Figure 6.2. The two panels have dimensions of 1.1×2.2 m each, resulting in a total SAs area of $14.5\,\mathrm{m}^2$, of which $10.32\,\mathrm{m}^2$ represent the effective cells area. Regarding the architecture, some precautions are to be considered. Firstly, cell protection is guaranteed by means of an integrated diode system. This solution consists of Germanium (Ge) substrate that allows to bypass the damaged cell, thus preventing hot spots and further issues. Moreover, due to the satellite's LEO, another protection is needed to prevent the erosion of parts exposed to atomic oxygen. This is addressed by pure finishing of metal parts [67].

- **Sequential Shunt Switching Regulator (S3R)**: The SAs power outcome is regulated through a Direct Energy Transfer (DET) solution that implements a shunt based on a S3R cell system, which enables dissipating the generated excess power. This choice allows to partially reduce the power losses which are the main drawback of the DET solution. Furthermore, it represents a cheaper solution with respect to the adoption of a Maximum Power Point Tracking (MPPT). More in depth, the S3R controls each solar cell and is capable of operating the cells in three different ways: Conduction mode, where the cell delivers all power from the SAs section to the battery bus; Shunt mode, in which the excess of current is absorbed and the power is dissipated; Switch mode, which is in charge of modulating the power delivered to the bus switching between the previous modes [69].
- **Power Control and Distribution Unit (PCDU)**: The cPCDU manufactured by Beyond Gravity is employed on-board the satellite [70]. The power coming from the SAs is fed into the PCDU, which works in synergy with the on-board computer to distribute the power where needed. When in sunlight, the SAs generate electrical energy, which is subsequently conveyed by the PCDU to the various subsystems and, if necessary, part of it is directed to recharge the battery. In eclipse and when needed, the PCDU is also tasked to switch from charging to drawing power from the battery to satisfy the on-board power demand. This electronic unit is mounted on the lateral panel of the platform, which is covered with OSRs, with the aim of promoting heat exchange and proper cooling.
- **Battery**: The battery constitutes the secondary electric power source for the spacecraft. Two 12s56p modules of Sony US18650 cells, based on Li-Ion technology, are installed in the satellite for a total capacity of 156 A h [71]. The battery is necessary to provide electrical power before the deployment of SAs during LEOP; when the SAs are not exposed to sunlight in eclipses; whenever the SAs are unable to provide the requested power. The latter condition occurs only in non-nominal situations, since the panels are significantly oversized, as previously mentioned. The choice of Li-Ion is preferred over NiCd or NiH_2 batteries since it ensures a higher energy density and a better resistance to charge and discharge cycles. Similarly to the PCDU, the battery is mounted on a side panel to facilitate its cooling.
- **LCL and FCL**: The Latching Current Limiter (LCL) is a protection mechanism that serves to disconnect a load from the circuit in the event of an overload and needs a telecommand to reconnect it. In the case of ADM-Aeolus, the LCL is placed before all the loads except those devoted to the TTMTC subsystem, which are instead preceded by Foldback Current Limiters (FCLs). This is probably due to the fact that a FCL reduces the current in case of an overload

and once this condition ends, the current is automatically restored to its nominal value [72]. This component is suited to precede very important loads, such as the TTMTC components, that shall communicate with the Earth at any time and under any conditions, as stated in system requirement COM-1 [3, Sec. 6.8.1].

- **Pyro Circuit**: The pyro circuit is tasked with deploying the solar panels through the activation of thermal knives once the satellite is separated from the launcher. The circuit is physically secured through safety switches isolating the pyro circuit until it is actively used, as stated in MEP-18 [3, Sec. 6.3.4].
- **BR and BU**: The buses distribute power either from the primary or secondary source to the loads. In ADM-Aeolus, two types of buses operate at different voltage levels: the 50 V Unregulated Bus (BU), supplied by the solar arrays and including the battery, performs the essential task of feeding ALADIN, the high-power payload. The remaining loads, less power demanding, are fed by a Regulated Bus (BR) operating at 28 V. The component responsible for voltage lowering is the PCDU.

6.1.3 Power requirements

The power required per subsystem throughout the mission duration is summarised in Table 6.1. The computation is based on the average power demand of each active component per subsystem during each phase and mode. The operational modes SCM, COM and CAL show high power requirements

S/S	**LEOP**	**Commissioning**	**Operational**					**EOL**
			SCM	**TCM**	**COM**	**CAL**	**SM**	
P/L [W]	0	840	840	0	840	840	0	0
PS [W]	0	0	0	112	0	0	0	112
TTMTC [W]	50	80	0	0	80	80	50	80
AOCS [W]	147	186	186	285	186	186	147	285
TCS [W]	200	200	200	200	200	200	200	200
EPS [W]	120	120	120	120	120	120	120	120
OBDH [W]	40	40	40	40	40	40	40	40
$\mathbf{P_{req}}$ [W]	557	1466	1386	757	1466	1466	557	837
20% mar. [W]	111.4	293.2	277.2	151.4	293.2	293.2	111.4	167.4
$\mathbf{P_{siz}}$ [W]	668.4	1759.2	1663.2	908.4	1759.2	1759.2	668.4	1004.4

Table 6.1: Power budget per mode.

since ALADIN is working continuously. Conversely, the power request decreases significantly in modes where the payload is inactive, like the TCM and the SM. Hence, the payload largely affects the overall power budget. The value reported for the PS accounts for the power required to heat the catalytic bed, estimated using typical values from [24], and to open the valves considering that four thrusters are active simultaneously [21]. The TTMTC power consumption changes throughout the mission phases. In particular, the highest power demand of 80 W is reached when scientific data are sent using the X-band antenna. While, when only S-band antennas are transmitting telemetry, the power requirement remains lower. Also for the AOCS the power demand varies depending on the mode, as already detailed in the dedicated report. In first approximation, the power consumption of the TCS is considered constant during all mission phases. This stems from the fact that ALADIN shall be maintained within strict temperature ranges even when it is not operative. The requested power for the EPS accounts for the internal consumption of the PCDU, which is approximately 5% of the input power [52]. This value is estimated considering the maximum possible input power of 2400 W and it is kept constant for a conservative approach. The OBDH remains active during all mission phases to process and store data. Typical values for its power demand are found in [24]. Finally, the total power request is increased to account for a margin of 20%, according to MAR-PWR-040 [12, Sec. 3.6], resulting in the sizing power.

6.1.4 EPS operational profile

The functioning of the EPS is strongly influenced by its orbit and mission profile. Periods of eclipse occur depending on the Earth's position relative to the Sun. More in depth, these eclipses occur close to the summer solstice, when the maximum duration of 25 minutes is reached. Since the satellite is launched on the 22$^{\text{nd}}$ of August, it experiences short moments of eclipse during LEOP that gradually decrease over time [71]. In addition, during CAL, the spacecraft performs the calibration of ALADIN on a weekly basis. During this procedure, the satellite goes from an inclination of 35° to a nadir pointing attitude, thus degrading the aspect angle. In the re-entry phase, the solar arrays are not exposed to the Sun during the orbit correction manoeuvres. This analysis, together with the required powers shown in Table 6.1, is important to identify the sizing condition of the EPS.

6.2 Electric Power Subsystem reverse sizing

In the present section, a reverse sizing of the ADM-Aeolus EPS subsystem is carried out and the capability of the solar arrays and battery to provide sufficient power throughout all mission phases and modes is proven.

6.2.1 Solar Arrays

In order to size the solar arrays, it is necessary to determine the power required by the satellite during sunlight and eclipse phases, considering the worst case scenario. In the ADM-Aeolus mission, the maximum amount of power required, as can be seen from Table 6.1, is associated to CAL during the Operational Phase, which also implies the worst aspect angle. This power requirement is unaffected by whether the satellite is in eclipse or sunlight, consequently $P_e = P_s = 1759.2$ W. In addition, the worst case scenario depends on the relative durations of the illuminated and shadowed orbital phases, which vary throughout the year, as previously mentioned in section 6.1.4. The sizing condition is associated to the summer solstice, where the duration of the eclipse is the longest: $T_e = 1475$ s and $T_s = 4075$ s [68]. Other additional data, related to the EPS architecture and the type of solar cells used [71], are shown in Table 6.2.

$\mathbf{X_e}$	$\mathbf{X_s}$	$\mathbf{P_0}$	ε	$\mathbf{I_D}$	**dpy**	**years**	$\mathbf{A_{cell}}$	$\mathbf{V_{sys}}$	$\mathbf{V_{cell}}$	$\mathbf{T_{orbit}}$
0.65	0.85	1365 W m^{-2}	0.27	0.8	0.03	3.25	30.18 cm^2	52.5 V	2.275 V	5550 s

Table 6.2: Data adopted for SA sizing.

The power that the solar arrays must be able to produce is computed through Eq (6.2.1).

$$P_{sa} = \frac{\left(\frac{P_e \cdot T_e}{X_e} + \frac{P_s \cdot T_s}{X_s}\right)}{T_s} \tag{6.2.1}$$

In order to calculate the total area needed for the solar arrays, it is necessary to find the power per unit of surface produced at the end of the mission. For this calculation, it is necessary to take into account the solar aspect angle variation that takes place during the mission. The variation of this angle along one orbit depends on the Earth's position with respect to the Sun. In this case, the cosine loss factor is computed taking into account the characteristics of the worst case scenario previously described. The procedure is derived from [68] and reported below.

$$a_{\text{cal}} = 0.6093, \quad b_{\text{cal}} = 0.3591, \quad \phi = \left(\frac{360 \cdot (T_{orbit} - T_e)}{2\,T_{orbit}} + 90\right), \quad c = \frac{2\pi}{T_{orbit}} \tag{6.2.2}$$

$$y(t) = a_{\mathrm{cal}} + b_{\mathrm{cal}} \cdot \cos(ct + \phi), \quad \mathrm{CLF} = \frac{1}{T_s} \int_0^{T_s} y(t)\, dt \tag{6.2.3}$$

The CLF parameter is computed as the mean value of the function y(t). The latter represents the cosine loss factor over the sunlight period and, through the parameters a_{cal}, b_{cal} and c, takes into account the inclination of the orbit, the solar array cant angle, the attitude of the satellite and the declination of the sun in that specific case [68]. The computed value is $CLF = 0.7241$.
At this point, it is possible to estimate the power per unit surface of the solar array at the beginning of life and thus at the end-of-life. A first approximation of the area of the solar arrays is then calculated and subsequently, considering the area of a single cell, a preliminary estimation of the number of cells is computed. After that, considering the voltage supplied to the system and that provided by a single cell, both reported in Table 6.2, the number of cells in series and in parallel can be calculated, taking into account one string failure (MAR-PWR-060 [12]). Finally, the exact values associated to the number of cells, the solar array area and the powers provided at the beginning and end of life are computed.
By analysing the results reported in Table 6.3, it is possible to notice that the calculated values are higher than those found in the literature [71]. The number of cells and therefore the necessary surface of the solar arrays is overestimated. This discrepancy can be mainly traced back to the partial inaccuracy of the power budget related to the worst case scenario, since some power values derive from estimates and approximations. Indeed, lower discrepancy is found when performing the same computations using the actual values of $P_{\mathrm{s}} = 1168.7\,\mathrm{W}$ and $P_{\mathrm{e}} = 1186\,\mathrm{W}$ [71], augmented by a 20% margin, according to the regulations (MAR-PWR-040 [12, Sec. 3.6]). The results, closer to the real ones discussed in section 6.1, are shown in the second row of Table 6.3.

Case	$\mathbf{P_s}$	$\mathbf{P_e}$	$\mathbf{N_{par}}$	$\mathbf{N_{ser}}$	$\mathbf{N_{cell}}$	$\mathbf{A_{sa}}$	$\mathbf{P_{BOL}}$	$\mathbf{P_{EOL}}$
Estimate	1759.2 W	1759.2 W	219	24	5256	15.9 m^2	3386.7 W	3067.5 W
Real	1402.4 W	1423.2 W	176	24	4224	12.7 m^2	2721.7 W	2465.2 W

Table 6.3: SA sizing results.

6.2.2 Battery

The satellite resorts to the energy stored in the battery both during eclipse and during LEOP, before the solar arrays are unfolded, as detailed in section 6.1.4. Both conditions are analysed, in order to verify the proper functioning of the system in the worst condition.
The first sizing condition is represented by the longest eclipse while the satellite is in Calibration Mode, which is the most power demanding. The two battery modules shall be able to store enough energy to provide the requested power for the whole duration of the eclipse. Eq (6.2.4) shows the energy stored in each battery pack to satisfy the requirements.

$$E_{\mathrm{batt}} = \frac{\frac{P_e\, T_e}{X_e}}{N_{\mathrm{batt}}\, \eta_{\mathrm{EOL}}\, DOD} \tag{6.2.4}$$

The number of battery cells in each string can then be computed as the ratio between the requested voltage and the voltage of each cell. Subsequently, the number of strings to be put in parallel is computed considering the overall requested energy and that produced by each string, accounting for the loss of one string, as mentioned in MAR-PWR-060 [12].

$$E_{\mathrm{string}} = \mu\, C_{\mathrm{cell}}\, (V_{\mathrm{cell}}\, N_{\mathrm{batt,s}}) \tag{6.2.5}$$

The overall power stored in a single battery and its capacity can finally be computed.

$$E_{\mathrm{batt,real}} = E_{\mathrm{string}}\, N_{\mathrm{batt,p}} \tag{6.2.6}$$

$$C_{\text{batt}} = \mu\, C_{\text{cell}}\, N_{\text{batt,p}} \tag{6.2.7}$$

The data used for the sizing and the results are shown in Table 6.4.

$\mathbf{N_{batt}}$	$\mathbf{V_{batt}}$	$\mathbf{C_{cell}}$	$\boldsymbol{\eta_{EOL}}$	**DOD**	$\boldsymbol{\mu}$	$\mathbf{N_{batt,s}}$	$\mathbf{N_{batt,p}}$	$\mathbf{E_{batt,real}}$	$\mathbf{C_{batt}}$
2	3.6 V	3 A h	0.8	0.3	0.8	15	19	2462.4 W h	45.6 A h

Table 6.4: Battery sizing data and results.

The values can be compared to the features of the actual battery installed on the satellite, described in section 6.1.2. It can be seen that there is a slight discrepancy in the number of battery cells in series, which might be caused by uncertainty in the knowledge of the actual voltage of the cells installed on the satellite. In contrast, the number of battery cells in parallel resulting from the reverse sizing is much smaller than that of the actual battery. This is probably due to the fact that the sizing condition for the battery capacity is not represented by an eclipse during Calibration Mode.
More likely, the most critical condition for the battery is represented by the LEOP. Therefore, it is necessary to prove that the battery is able to provide the required energy for enough time, in order to allow the correct deployment of the solar arrays. This is done using a reverse sizing approach, considering the actual battery installed on the satellite and computing the related survival time. Eq (6.2.5) and Eq (6.2.6) are used to calculate the amount of energy stored in each battery unit. Then, Eq (6.2.8) is used to compute the maximum operational time during LEOP, considering the correspondant power shown in Table 6.1. It is important to note that the equation differs from Eq (6.2.4) of a factor η_{EOL}, since the LEOP occurs in beginning-of-life condition for the batteries. The efficiency values adopted are those used in the previous sections.

$$T_{\text{LEOP,max}} = E_{\text{batt}}\, N_{\text{batt}}\, DOD\, \frac{X_e}{P_e} \tag{6.2.8}$$

The resulting value is $T_{\text{LEOP,max}} = 3\,\text{h}\,20\,\text{min}$. According to literature data regarding similar missions [73] [74], the deployment of the solar arrays occurs within one hour from the separation of the satellite from the launcher's upper stage. The value found has to be considered as an upper limit and was likely designed according to broad margins. Indeed, it is aligned with those related to similar missions.

6.2.3 Mass, volume and data budget

The mass and volume budget of the SAs is taken from [67], considering an approximate thickness of 5 cm. Battery mass and volume are estimated starting from the data in section 6.1.2, considering the specific energy density values of $206.9\,\text{W}\,\text{h}\,\text{kg}^{-1}$ and $500\,\text{W}\,\text{h}\,\text{L}^{-1}$. Regarding the PCDU, in absence of specific information related to the ADM-Aeolus mission, the mass and volume of the Airbus LEO PCDU EVO module is taken as a reference [75]. Given that the latter provides up to 4 kW, which is higher but comparable to that of the actual mission, it can reasonably be used for a first analysis. The overall data budget for the EPS, reported per sampling, is derived from [24, Table 16-13].

Quantity	**SAs**	**PCDU**	**Battery**	**Total**
Mass [kg]	19.6	16.5	37.7	73.8
Volume [L]	72.5	18.5	15.6	106.6
Data [kbit]	81.9			81.9

Table 6.5: EPS mass, volume and data budget.

Chapter 7

On-Board Data Handling Subsystem

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7.1 On-Board Data Handling Subsystem overview

7.1.1 Introduction

The OBDH subsystem is responsible for managing and processing data exchanged with sensors, instruments, components and, more in general, all on-board subsystems. Specifically, it handles the control of the payload and of the platform, including its attitude, power, propulsion and communication. Regarding the last point, the OBDH subsystem is also responsible for acquiring, storing, and distributing to the downlink facility the data generated by both ALADIN and the platform, as specified in requirement DHS-6 [3, Sec. 6.7.2].

7.1.2 OBDH architecture

In this section the main components of the OBDH subsystem are discussed, starting from the Control and data management unit (CDMU), with a focus on its architecture and communication interfaces. Produced by RUAG, it represents the heart of the satellite's OBDH subsystem, serving as the fulcrum for managing data and commands. As shown in Figure 7.1, it adopts a bus architecture that offers advantages such as improved scalability, reduced wiring complexity, and easier fault detection. The Processor Module (PM) interfaces with two distinct MIL-STD-1553 buses, i.e., one Internal Control Bus (ICB), and one that connects the CDMU to external units, represented by AOCS units, PCDU and ALADIN Control & Data Management (ACDM) unit. Furthermore, the connection with the AST is implemented using a different bus, specifically the RS422 High Speed Universal Asynchronous Receiver Transmitter (HSUART). As reported in [8], the ADM-Aeolus on-board software is required to manage the mission timeline while ensuring sufficient data storage capacity to support the 7-day orbit repeat cycle. This top level requirement directly impacts the sizing of the OBDH hardware, particularly the memory and data handling capabilities of the CDMU, which must be carefully sized to prevent data loss. Each component of the CDMU, as well as the adopted buses and the ACDM, are described hereafter.

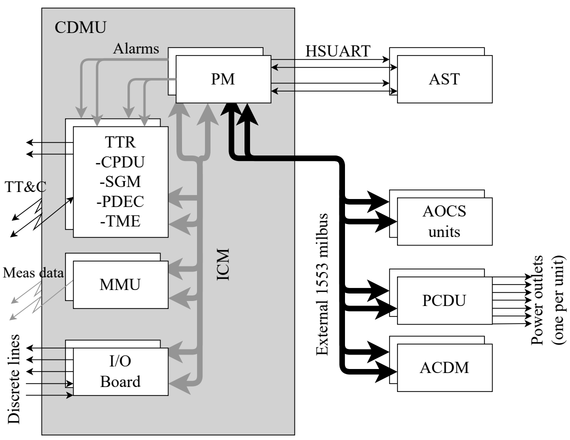

Figure 7.1: OBDH overview [76].

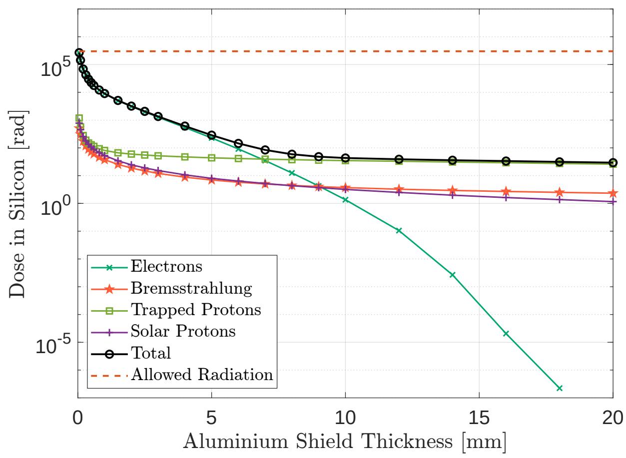

Figure 7.2: SPENVIS analysis.

- **Processor Module (PM)**: The processor is an ERC-32, a radiation tolerant version of the SPARC V-7 processor [77]. The PM executes 10 MIPS and 2 MFLOPS [78]. It is supported with 6 MB of RAM and a PROM of up to 16 MB. The relatively small RAM size is probably related to the fact that ALADIN has its own processing unit, which offloads computational burden from the CDMU. The whole module is cold redundant. The radiation hardness is tested up to 300 krad, which is consistent with the level of expected radiations in LEO along the 3.25 years duration of the mission without further shielding, as depicted in Figure 7.2. The latter represents the SPENVIS analysis considering the actual levels of solar activity, which remain low for most of the mission [13].

- **Telemetry, Telecommand and Reconfiguration (TTR)**: The TTR performs several crucial functions within the OBDH, such as decoding remote controls, managing telemetry encoding, preserving essential data during reconfigurations, and executing high-priority commands in the event of software failures. These features are implemented through four main hardware elements:
  - **Command Pulse Distribution Unit (CPDU)**: The CPDU manages and executes high priority commands directly on the onboard hardware, bypassing the software. By operating at the hardware level, the CPDU can execute commands even in situations where the onboard software is functioning incorrectly or is inoperative. This ensures the ability to control critical satellite functions in emergency conditions. For example, in the event of a failure of the on-board software, the CPDU can be used to restart essential subsystems or change the operational state of the satellite [79].
  - **Safe Guard Memory (SGM)**: The SGM is a permanently powered non-volatile memory. It is used to retain data during reconfigurations or reinitializations of the PM. At the same time, the two software images of the PM, one for nominal mode and one for safe mode, are saved in this memory to restore the system [4]. Unlike a ROM, which cannot be updated, a SGM allows for in-orbit reprogramming and system recovery in case of faults. This bestows greater robustness and adaptability, making the SGM a better solution for OBDH.
  - **Packet Decoder (PDEC)**: The PDEC is used for the decoding of telecommands [80]. This component is of utmost importance and is operated in hot redundancy, since decoding TC is a vital functionality of the spacecraft [8].
  - **Telemetry Encoder (TME)**: The TME manages the encoding of telemetry data collected by the various subsystems and prepares their transmission to the ground stations.

  The reconfiguration functionality of the TTR is significantly connected to the FDIR and is carried out by the Reconfiguration Modules (RMs), which monitor autonomously any alarm signal coming from the PM or the Application software (APSW) and perform the reconfiguration of the PM accordingly. If appropriate, the RMs can switch the CDMU into safe mode [76].
- **Mass Memory Unit (MMU)**: The MMU is a solid-state memory designed to store critical mission data, such as scientific measurements, telemetry, and command sequences [81]. It provides a capacity of 8 GB of on-board data storage [7], and is organized in several independent circular buffers that continuously record both housekeeping and scientific telemetry [8].
- **Input/Output (I/O) Board**: Its main function is to provide digital and analogue interfaces between the on-board computer and the various subsystems. It allows the CDMU to directly monitor and control devices that do not communicate via a digital bus.
- **Bus:** ADM-Aeolus implements two MIL-STD-1553 buses, widely used in space applications for their robustness and resistance to disturbances. However, given the limited data rate of this type of bus, which has an upper limit of $1\,\mathrm{Mbit\,s^{-1}}$, a different bus is adopted for the AST, where robustness must be accompanied by rapid response and low latency. The selected solution is RS422 HSUART [76], a bus that likewise remains quite typical in space applications. An additional advantage of using a dedicated interface for the AST is that it isolates this sensor, fundamental for attitude determination, from the traffic of a shared communication bus. Both buses are implemented with cold redundant couples [76].
- **ALADIN Control & Data Management (ACDM):** The ACDM, which is part of ALADIN and supervises all its activities, interfaces with the CDMU. It provides full autonomy in operation management, ensures the synchronization between laser emission and backscatter signal acquisition, the data processing and data stretching towards the spacecraft and the thermal regulation functions as seen in similar LIDAR implementations [82].

7.1.3 Software and autonomy

From the software point of view, ADM-Aeolus implements high autonomy. This includes a FDIR system integrated in the APSW, the CDMU software, forming an essential part of the OBDH subsystem [8]. Practically, this autonomy is achieved via a weekly-uploaded command schedule from ground, structured either as a MTL (time-driven) or an OPS (orbit-based). In both cases the CDMU saves these commands and delivers them to the subsystems at the correct time. The FDIR minimizes ground intervention during failure scenarios. Two main FDIR strategies are adopted: a **fail-operational**, implemented for non-critical failures, in which operations continue during fault recovery (e.g., X-band transmitter failure); and a **fail-safe** one, where operations are suspended until the fault is corrected (e.g., ALADIN failure). The FDIR concept is built around a top-down onboard control architecture relying also on hardware [8]:

- **Top level**: TTR boards operated in hot redundancy within the CDMU contain the RMs which oversee the health and function of the CDMU itself.
- **Intermediate level**: APSW monitors and controls the spacecraft units through telemetry data and sends telemetry packets to ground. Dumping of on-board telemetry is started autonomously by the APSW using the OPS schedule and on-orbit position data provided by the GPS sensor. This data dumps are fundamental to ensure that no information is overwritten.
- **Lowest level**: some units perform their own built-in health checks and report it through the telemetry to the CDMU software [8].

7.2 On-board memory sizing

A first estimate of the data budget is shown in Table 7.1. For each subsystem, the most relevant parameters are proposed, and for each of them a consistent size and sampling frequency are established. During its operational phase, the satellite downlinks both telemetry and payload data approximately once per orbit. Therefore, the data volume is computed considering the duration of one orbit, that is 5550 s, for almost all quantities. The data volume for the transmitted and received packet counters in the TTMTC is computed considering an average link time of 600 s. Finally, concerning the AOCS, the MTQ torque is considered to be of interest only during the desaturation of reaction wheels, that lasts approximately 100 s.

The payload section includes both scientific measurements and data regarding the components of ALADIN. The scientific data refer only to the raw measurements and amount to 50 MB [1]. This is a relatively low volume since most of the processing is performed on ground, where greater computational capacity is provided in order to send refined data to users in near-real-time. Data regarding the health status of ALADIN, reported in Table 7.1, concern only its main components. It is reasonable to assume that this value is largely underestimated given the high complexity of the payload which includes sophisticated components that need to be monitored thoroughly. Also, since one of the mission objectives is to demonstrate the effectiveness of the DWL technology, many data may need to be downlinked to assess the performance of ALADIN.

Observing all the subsystems, it is clear that current, voltage and temperature are among the most relevant quantities of the components, as they provide essential information about their correct functioning and are easily measurable. The parameters with the highest storage frequency are, as can be seen, those related to quantities with great variability, such as position, angular rates, attitude and pointing error. These data, that are mainly associated with the AOCS, are of great interest, given that pointing accuracy is a mission driver.

From Table 7.2, it can be seen that the total amount of data stored in an orbit is 366.21 MB, accounting for a 400% margin, deemed appropriate for a preliminary design. Nonetheless, this value is much lower than the capacity of the solid state memory. This is due to the limited knowledge about the required stored quantities. Moreover, oversizing the onboard memory has minimal impact on mass, volume, and power consumption.

7.3 Data budget

S/S	Parameter	n.	Data type	Data size [bit]	Freq. [Hz]	Data rate [bit s^{-1}]	Data volume [kbit]
P/L	Raw measurement data	N.A.	N.A.	N.A.	N.A.	N.A.	409600
	ALADIN mode	1	uint8	8	0.1	0.8	4.34
	PLH status	2	bool	8	e.b.[1]	0	0
	PLH temperature	2	sint16	16	0.5	16	86.72
	RLH status	2	bool	8	e.b.[7.1]	0	0
	RLH temperature	2	sint16	16	0.5	16	86.72
	TLE current	1	sint16	16	1	16	86.72
	TLE temperature	1	sint16	16	0.5	8	43.36
	Primary mirror temperature	12	sint16	16	0.5	96	520.31
	DFU temperature	2	sint16	16	0.5	16	86.72
	DFU current	2	sint16	16	1	32	173.44
PS	Tank temperature	2	sint16	16	0.5	16	86.72
	Tank pressure	2	uint16	16	0.5	16	86.72
	Valve status	10	uint8	8	0.5	40	216.80
TTMTC	TTMTC mode	1	uint8	8	0.1	0.8	4.34
	Receiver current	2	sint16	16	1	32	173.44
	Transmitter current	3	sint16	16	1	48	260.16
	S-band status (RX/TX)	2	bool	8	0.1	1.6	8.67
	S-band antenna temperature	2	sint16	16	0.5	16	86.72
	X-band antenna temperature	1	sint16	16	0.5	8	43.36
	Amplifier current	2	sint16	16	1	32	173.44
	Transmitted packet counter	2	uint8	8	10	160	93.75
	Received packet counter	1	uint8	8	10	80	46.88
AOCS	AOCS mode	1	uint8	8	0.1	0.8	4.34
	Reset count AOCS	1	uint8	8	e.b.[7.1]	0	0
	Orbit propagation	6	float32	32	10	1920	10406.25
	Estimated Quaternion	4	float32	32	10	1280	6937.50
	Estimated angular rate	3	float32	32	100	9600	52031.25
	Pointing error	1	float32	32	10	320	1734.38
	Rate error	3	float32	32	10	960	5203.13
	AST output	4	float32	32	10	1280	6937.50
	AST temperature	1	sint16	16	0.5	8	43.36
	IMU voltage	1	sint16	16	1	16	86.72
	IMU temperature	1	sint16	16	1	16	86.72
	IMU current	1	sint16	16	0.5	8	43.36
	IMU output	4	float32	32	100	12800	69375.00
	MTQ current	3	sint16	16	1	48	260.16
	MTQ torque	3	sint16	16	10	480	46.88
	MTQ status	1	bool	8	e.b.[7.1]	0	0
	RW Torque	4	sint16	16	10	640	3468.75
	RW Speed	4	sint16	16	10	640	3468.75
	RW saturation flag	4	bool	8	e.b.[7.1]	0	0

Continues on next page

[1] Event-based.

S/S	Parameter	n.	Data type	Data size [bit]	Freq. [Hz]	Data rate [bit s^{-1}]	Data volume [kbit]
TCS	Radiator temperature	6	sint16	16	0.5	48	260.16
	Cold Plate temperature	4	sint16	16	0.5	32	173.44
	Heater current	20	sint16	16	1	320	1734.38
EPS	EPS status	1	uint8	8	0.1	0.8	4.34
	PCDU temperature	1	sint16	16	0.5	8	43.36
	Battery temperature	2	sint16	16	0.5	16	86.72
	Battery inward current	2	uint16	16	1	32	173.44
	Battery outward current	2	uint16	16	1	32	173.44
	Battery voltage	2	sint16	16	1	32	173.44
	Battery number of cycles	1	uint16	16	e.b.[7.1]	0	0
	P/L bus voltage	1	sint16	16	1	16	86.72
	System bus voltage	1	sint16	16	1	16	86.72
	SA strings current	38	sint16	16	1	608	3295.31
	SA panel temperature	6	sint16	16	0.5	48	260.16
	S3R strings voltage	38	sint16	16	1	608	3295.31
OBDH	CDMU voltage	1	sint16	16	1	16	86.72
	CDMU time	1	uint32	32	100	3200	17343.75
	CDMU watchdog	1	uint16	16	1	16	86.72
	CDMU current	1	sint16	16	1	16	86.72
	CDMU stored packet counter	1	uint16	16	0.1	1.6	8.67
	CDMU temperature	1	sint16	16	0.5	8	43.36
	CDMU boot counter	1	uint8	8	e.b.[7.1]	0	0
	CDMU boot cause	1	uint16	16	e.b.[7.1]	0	0
	CDMU usage	1	uint8	8	1	8	43.36
	Bus status	2	bool	8	e.b.[7.1]	0	0
	ACDM voltage	1	sint16	16	1	16	86.72
	ACDM current	1	sint16	16	1	16	86.72
	ACDM watchdog	1	uint16	16	1	16	86.72
	ACDM temperature	1	sint16	16	0.5	8	43.36
	ACDM stored packet counter	1	uint16	16	0.1	1.6	8.67
	ACDM usage	1	uint8	8	1	8	43.36
	ACDM boot counter	1	uint8	8	e.b.[7.1]	0	0
	ACDM boot cause	1	uint16	16	e.b.[7.1]	0	0

Table 7.1: Data budget table.

Parameter	Value
Total data volume per orbit [Mbit]	585.94
Margin 400% [Mbit]	2343.77
Margined data volume per orbit [Mbit]	2929.71
Margined data volume per orbit [MB]	366.21

Table 7.2: Total data volume per orbit.

Chapter 8

Structure and Configuration

Contents

Change log	
Issue	*Comments*
1	First release

8.1 Structure and configuration

8.1.1 Introduction

The present report analyses the structure and configuration of ADM-Aeolus. The focus will be on the launcher selection, the interfaces, the structural solutions adopted, and the SARM. Additionally, the layout of the components will be discussed.

8.1.2 Launcher selection

The satellite must be designed to ensure compatibility in terms of mass, volume, and structural loads with at least two different launchers, as specified in the requirement GSR-4 [3, Sec. 6.1]. This requirement allows greater flexibility in launcher selection during the design's final stages. ADM-Aeolus was launched with the Vega launcher operated by Arianespace. Vega was an optimal choice, as the mission profile matched its operational capabilities [83]. Compatibility was also verified structurally [84]. The structural analysis had to consider the four stages of the launcher (three solid-propellant stages and the AVUM liquid upper stage), as well as the axial and lateral loads during ascent. Stage separations introduce shocks and vibrations, requiring FEM analysis. Furthermore, due to the solid propulsion system, the structural analysis must take into account cyclic loads caused by pressure oscillations in the combustion chambers. These oscillations can couple with the satellite's own structural modes and require careful dynamic verification to ensure the protection of sensitive instruments like ALADIN.

8.1.3 Structure

ADM-Aeolus is based on a compact and robust structural configuration designed to transmit and sustain all possible loads. Since the launch environment is the most demanding in terms of loads, it is reasonable that the sizing of the structure is done accordingly. From the structural viewpoint, two modules, connected through an interface structure of octagonal shape [85], are found: the platform, composed of a primary, secondary and tertiary structure, and ALADIN, which includes a primary and secondary structure.

8.1.3.1 Platform structure

The **primary structure** of the platform shall support most of the loads and ensure satellite rigidity. It derives from a past mission, Mars Express. This choice is probably linked to advantages such as lower cost, faster development, proven reliability, and flexibility to modification. It has a box shape with dimensions $1.9 \times 1.64 \times 1.6$ m and is made of aluminium honeycomb sandwich panels. Specifically, these panels are made of honeycomb (generally type 1/8-5056-0.001P) 10-20 mm thick, bonded to aluminium facesheets of thickness ranging from 0.2 to 0.3 mm and up to 0.5 mm additional doubler for local reinforcements [86]. In addition, three more transversal walls can be found inside the cube, two orthogonal to the Z direction and one orthogonal to the X direction, which act as additional structural supports and mounting surfaces, as depicted in Figures 8.1, 8.2. Aluminium is chosen for its light weight and good mechanical and thermal properties, while the technology adopted in the panels is justified by its heritage in the space sector. Additionally, the ring adapter is considered part of the primary structure, since it is the main load path from the spacecraft body to the launcher interface.
The **secondary structure** includes the support structure for solar panels. Probably made of aluminium, it supports the operational loads resulting from in-flight manoeuvres and ensures the correct orientation of the solar panels. The **tertiary structure** of ADM-Aeolus includes local support elements such as mounting brackets for the two propellant tanks and structural devices for thrusters, CESSs and antennas. These components are not part of the primary load path, but are essential to ensure proper alignment, vibration isolation, and mechanical integration. Their design focuses on minimizing mass, while maintaining sufficient rigidity and thermal compatibility with the surrounding structure.

Regarding the mass of the platform structure, since no data is available in the literature, a broad estimation is performed. Considering the dimensions of the platform, its composition and the densities of aluminium honeycomb ($72.1\,\mathrm{kg\,m^{-3}}$ [87]) and facesheets ($2680\,\mathrm{kg\,m^{-3}}$), the computed mass is 108 kg.

8.1.3.2 ALADIN structure

The **primary structure** of ALADIN consists of the main structure, the satellite interface and the star tracker support. The design of such structures is compliant with three main drivers: withstand very high loads, convey maximum stiffness to decouple ALADIN from the satellite's bus, and provide high stability despite inhomogeneous thermal distributions. To meet these strict requirements, the structure is made of sandwich panels with CFRP face-sheets and aluminium honeycomb cores [88]. This configuration offers both strength and light-weight properties [89]. Additionally, the panels have low CTE and CME to minimize deformations due to thermal gradients. Finally, the interface includes three isostatic bipods [4] that are able to compensate both gravitational and thermal deformations, thus isolating the optical components and keeping the mirror's optical axis in place.
The **secondary structures** include the large external baffle, three thermal heat shields and a thermal hood. In this case the design aims to build structures able to withstand loads caused by the dynamic excitation of their own masses [88]. To achieve this, the external baffle framework is made using CFRP struts and aluminium rings and nodes. CFRP is selected for its high strength-to-weight ratio while aluminium is a suitable choice for rings and joints thanks to its optimal machinability. The baffle framework configuration, which has a lower conical part, is optimized to have maximum stiffness. This property is essential for the housing of antennas and of CESSs, located on top of the baffle, where high accelerations can be reached. To further mitigate the effects induced by dynamic accelerations, ultra light-weighted brackets are used for the antennas. Since the bondings join materials with different CTEs and the external baffle experiences significant temperature variations, a specific strut end design is introduced to minimize stresses. It is also worth noting that the baffle is shortened on the anti-Sun direction of the satellite to reduce mass and air drag [90]. The other secondary structures refer to thermal control for optical components. Specifically, thermal heat shields provide thermal protection to the primary mirror, whereas the thermal hood covers the optical bench assembly. These choices highlight the importance of ensuring a constant thermal environment to ALADIN.
The ALADIN structure has a mass budget of 140 kg [91, Sec. 30.10.5]. The total mass, including that of the platform, is 248 kg, corresponding to 22% of the dry mass, consistent with values found in [24].

8.1.3.3 Solar Array Rotation Mechanism

The only components that need to be deployed after launch are SAs, so a dedicated mechanism is mounted on-board. In particular, two mirror SARM units are required to rotate the SAs in opposite directions. Each unit includes a MHRM and a FRED panel hinge, two space-qualified components already present in the market, selected to keep the design simple and cost effective [92]. During launch, the MHRM secures the SAs to the satellite structure using a Dyneema hold-down cable. Once in orbit, the release mechanism is activated using two thermal knives, one nominal and one redundant. Indeed, MHRM must be highly reliable since a failure in the deployment of the SAs would jeopardize the mission. After the hold-down cable is cut, the FRED panel hinge rotates the SAs exploiting two springs. Then two latches lock the hinge in its final configuration. It is crucial that the springs supply enough torque to overcome resistances, especially the retarding torque caused by the cable harness. To minimize the deployment shock, the adopted solution involves placing the cable harness around a cylinder so that it remains flat and sufficiently long.

8.1.4 Configuration

In this section the configuration of the main components aboard ADM-Aeolus is described, based on renderings available on ESA's website [93][94][95].

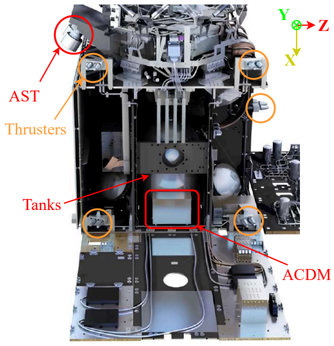

Figure 8.1: X-Z plane.

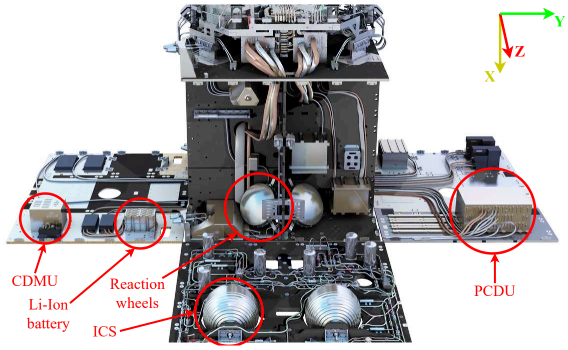

Figure 8.2: X-Y plane.

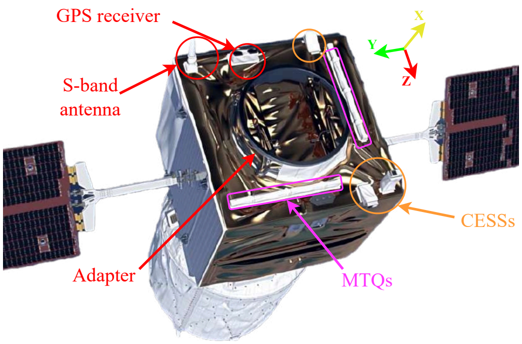

Figure 8.3: Top view.

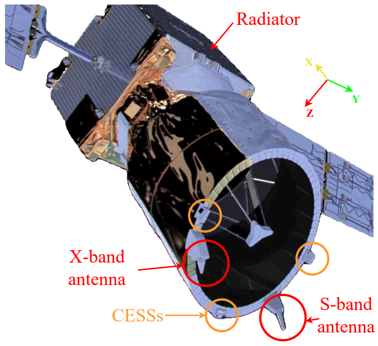

Figure 8.4: Bottom view.

8.1.4.1 Internal components

The tanks, when full, are the heaviest components of the entire satellite and, together with other massive units such as the Li-Ion battery, the PCDU, the CDMU, and the In-Situ Cleaning System (ICS) [96], play a crucial role in defining the overall centre of mass. To ensure proper mass distribution and maintain the centre of gravity aligned with the ALADIN optical axis, these elements are strategically arranged across the platform. In particular, since the other components are located towards the +Z face of the satellite, the **tanks** are positioned slightly in the -Z direction. Placing the centre of gravity along the ALADIN LOS facilitates the control of its pointing, as it minimizes dynamic coupling effects during attitude control. Moreover, the **ICS**, used for cleaning the optical path of ALADIN, is placed close to the instrument, with the feeding line directed towards it for functional purposes.
The **IMU** is located on the face oriented along the Z axis and is divided into two main components: the sensor head, positioned near the ALADIN to accurately measure angular rates in proximity of the payload, and the electronics unit, which is instead placed closer to the external surface to allow heat dissipation and to avoid interference with the sensor. Concerning the **RMU**, no information regarding its location is available in the literature. Nevertheless, its lack of strict thermal or operational constraints makes it flexible in placement.
The **RWU** is placed near the centre of mass to minimize the effects of the vibrations it induces, arranged in a pyramidal configuration symmetrical with respect to ALADIN axis. This setup ensures a preferential direction of operation, which, for aiming requirements, is aligned with the LOS of ALADIN. **MTM** are also part of the internal components of ADM-Aeolus. In contrast to what is usually done for positioning this component, they are placed inside the satellite since they are used to give only rough estimates on the satellite's attitude during initial acquisition and safe mode.
Finally, the **battery**, **CDMU** and **ACDM** units are mounted on the panel opposite to the one on which resides the **PCDU**. These panels, located along the Y axis, are those with the white coating and therefore exhibit high emissivity. This choice is adopted to aid the thermal control.

8.1.4.2 External components

Most attitude sensors, as well as the payload and the antennas, are located on external surfaces of the platform, due to their requirements related to pointing, thermal control and illumination conditions.
ALADIN occupies the whole -X face of the platform. Its pointing drives the whole configuration design, since it determines the spacecraft's attitude and thus the orientation of all the other surfaces of the satellite. The size of the telescope, which is its biggest element, is limited to ensure compliance with the volume requirements from the launchers [7, Sec. 4.3.1].
The two **ASTs** are installed directly on the ALADIN structure to minimize misalignment errors [7, Sec. 4.3.1]. More specifically, they are placed on the -Z face, which is in shadow during nominal operations and are tilted on the XZ plane in order for them to be pointed towards the Zenith direction. Such configuration is integral to the proper functioning of this device, since it is very sensible to light and its performances are degraded or it could be damaged by exposure to the Sun or Earth's albedo.
The eight **CESSs** feature broad FOV to ensure full sky coverage. Therefore, they shall not be obstructed by long structures such as the ALADIN telescope (-X direction) or the solar arrays (±Y direction). To achieve so, three of the sensors are mounted at the end of the ALADIN telescope. This configuration allows the two sensors pointed towards the ±Y direction to be at the greatest possible distance from the SAs, thereby limiting FOV obstruction. Three other CESSs are on the +X faces. It is relevant to note that the one pointed towards the +X direction is located at the top of a support, which is likely necessary to avoid FOV obstruction due to the ring adapter and, before their deployment, by the SAs. Two other CESSs are located on the ends of the solar panels.
The three **MTQs** are aligned with the satellite's body axes. Those aligned with the Y and Z directions are located externally, on the +X face of the platform. This causes them to be continuously exposed to sunlight during nominal operations, which is not a problem thanks to their wide operative temperature range. The third MTQ, aligned with the X direction, is placed internally. The location of the three actuators allows them to not interfere with electronic components that might be sensible to strong magnetic disturbances, such as the AST and the P/L.
The two **GPS receivers** are located on the +X face of the satellite. They are tilted on the XZ plane in order for them to be approximately Zenith pointing during nominal operation and not be obstructed by the spacecraft body, thus achieving optimal sky coverage.
One of the **S-band antennas** and the **X-band antenna** are located on the ALADIN telescope. They are tilted on the XZ plane to be Zenith pointing in nominal condition. Another S-band antenna is located on the +X face of the satellite and is tilted on the same plane as the other two so as to be pointed in the opposite direction with respect to its counterpart. This configuration allows optimal sky coverage and SNR, both in nominal and failure scenarios.
The two **solar arrays** consist of long appendages exiting from the ±Y faces. In the stowed configuration they are folded and cover almost completely the face, partially obstructing the thrusters (which are not used before the SA deployment). In the deployed configuration, they are rotated by 45° in order to ensure maximum Sun aspect angle across the orbit [68].
Eight **hydrazine thrusters** are placed in pairs on the corners of the -Y face of the platform. They are symmetrically tilted by 15° on the YZ plane, thus reducing heat and particle flux on SAs [20], in order to avoid damage and reduction in performance.
The ALADIN **radiator** occupies the whole -Z face of the satellite. This solution allows it to be oriented towards deep space during nominal operations. It also allows the radiator to be in a face adjacent to the payload, in order to optimize the heat pipes' path. Actually, the selected face is the only one that allows to satisfy both requirements considering the satellite's attitude.
Finally, the **launcher adapter** is located on the +X face of the platform, opposite to ALADIN. This choice is mandatory due to the big size of the payload, in order for the satellite to correctly fit inside the launcher fairing.

Chapter 9

Mass Budget

Contents

9.1 Data retrieved from literature

S/S	**Mass** [kg]
P/L	450
PS	33.4
TTMTC	
Antennas	0.88
Other	N/A
AOCS	
RWU	26.8
MTQ	23.4
AST	6
IMU	12.7
RMU	0.18
CESS	0.64
MTM	0.6
GPS	7.2
Other	N/A
TCS	N/A
EPS	
PCDU	16.5
PVA	19.6
Other	N/A
OBDH	N/A
STR	N/A
Total dry mass (w/o adapter)	**1100**
Propellant	266
Total wet mass (w/o adapter)	**1366**
Adapter	77
Total launch mass (w/ adapter)	**1443**
Maximum launchable mass	**1500**

Table 9.1: Mass budget from literature data.

9.2 Data from reverse engineering

S/S	**Mass** [kg]	**Margin** [%]	**Mass with margin** [kg]
P/L	450		**450**
PS	13.9	5	**14.6**
TTMTC	44	5	**46.2**
AOCS			
RWU	26.8	5	28.1
MTQ	23.4	5	24.6
AST	6	5	6.3
IMU	12.7	5	13.3
RMU	0.18	5	0.19
CESS	0.6	5	0.7
MTM	0.6	5	0.7
GPS	7.2	5	7.6
Subtotal			**81.4**
TCS			
Radiator	7	10	7.7
Heat pipes	6	10	6.6
Cold plate	10	10	11
MLI	13	10	14.3
Beta cloth	3	10	3.3
OSR	1.4	10	1.5
Subtotal			**44.4**
EPS			
Battery	37.7	5	39.6
PVA	19.6	5	20.6
PCDU	16.5	5	17.3
Subtotal			**77.5**
OBDH	44	5	**46.2**
STR	108	20	**129.6**
Nominal dry mass (w/o adapter)	**851.6**		**889.9**
System Level Margin		20	
Total dry mass (w/o adapter)			**1067.9**
Propellant	217.1	20	260.5
Total wet mass (w/o adapter)			**1328.4**
Adapter	77	5	80.9
Total launch mass (w/ adapter)			**1409.3**
Maximum launchable mass			**1500**

Table 9.2: Mass budget from reverse engineering study.

9.3 Comments

Table 9.1 and Table 9.2 show the mass-breakdown budgets for the ADM-Aeolus mission. Specifically, the first one contains data taken from literature, while the second one includes data derived from the reverse engineering processes of previous assignments.
As can be seen in the first table, the data available in the literature are not exhaustive or complete. Precise information regarding the weight of the structure, the TCS, and the OBDH subsystem is not available. Therefore, no mass values are included in the corresponding rows.
For the TTMTC subsystem, AOCS, and EPS, the masses of the specific components are retrieved from the datasheets, while the total mass of each subsystem is not explicitly provided. Any additional masses, such as those related to cables and valves, are considered unknown and included in the "other" category. Instead, the payload mass, the propulsion subsystem mass, the total dry mass, the total wet mass, the adapter mass, and the total launch mass are well defined. Furthermore, given the characteristics of the Vega launcher, the maximum possible launch mass is also known.
Regarding Table 9.2, several strategies are adopted in the reverse engineering processes to define the mass budget. The payload mass is assumed to be known a priori and is therefore identical to that in Table 9.1, equal to 450 kg; no margin is considered. The mass of the propulsion subsystem is computed using simplified formulas. The tanks are sized considering an operating pressure of 2.4 MPa, rather than a burst pressure of 4.8 MPa. This explains why the predicted mass, even considering the margins, is much lower than the one found in the literature. For the TTMTC, the mass is instead approximated by taking into account missions similar to ADM-Aeolus (meteorology satellites in LEO), where it accounts for approximately 4-6% of the dry mass. The discrepancy between the values in Table 9.1 and Table 9.2 is clearly due to the lack of specific information regarding the ADM-Aeolus mission. In the AOCS, the mass is approximated by taking into account the relevant components and using reference values obtained from the literature. The difference between Table 9.1 and Table 9.2 therefore lies in the addition of margins in the second case. For the TCS, the mass of the components is calculated based on the material densities and dimensions involved. Although real reference data are unavailable for direct comparison, the fact that the TCS is approximately 4% of the total dry mass demonstrates that the estimated value is likely to be a realistic approximation. In the EPS, literature data are used for PVA and PCDU, while the battery mass is computed through the dedicated sizing. Again, a comparison with Table 9.1 cannot be made due to the lack of data. The OBDH mass is approximated considering similar missions, while the mass of the platform structure is evaluated considering its dimensions and densities. Direct comparisons with literature data are not possible.
The propellant mass is calculated starting from an estimated Δv budget of $406.44\,\mathrm{m\,s^{-1}}$, resulting from mission analysis, which provides a value of 217.1 kg. A 20% margin is then adopted, according to system requirement AOS-20 [3]. The output value is 260.5 kg, which is consistent with the actual mass of 266 kg.
A 5% Design Maturity Margin (DMM) is considered for the majority of subsystems, since most components are off-the-shelf. Instead, a DMM of 10% is adopted for the TCS to consider the uncertainty in the knowledge of the actual materials used on-board the satellite. A 20% margin is adopted for the structure, in order to account for the lack of knowledge on the density of the materials and the dimensions, and the mass of secondary structure elements which are not considered in the calculations. Adding all the contributions together and considering a system level margin of 20%, a total dry mass of 1067.9 kg is found. This value is close to the actual value of 1100 kg. Finally, considering the propellant mass, a total wet mass of 1328.4 kg is found, also consistent with the actual values. The total launch mass, including the adapter mass of 77 kg [97] and its relative margin, is therefore 1409.3 kg, less than the maximum limit of 1500 kg [83], justifying the compatibility of the satellite with the Vega launcher.

Nomenclature

α_{int} Internal face absorptivity

η_{EOL} End of life efficiency

μ Pack efficiency

σ Stefan-Boltzmann constant $[\mathrm{W\,m^{-2}\,K^{-4}}]$

ε Emissivity

ε Solar cells efficiency

ε_{int} Internal face emissivity

ε_{rad} Radiator emissivity

A Area $[\mathrm{m^2}]$

A_{cell} Solar cell area

A_{rad} Radiator area $[\mathrm{m^2}]$

$A_{c_{i\to j}}$ Conductive area $[\mathrm{m^2}]$

C_{batt} Total electric charge

C_{cell} Cell electric charge

CLF Cosine loss factor

DOD Depth of discharge

dpy Yearly degradation factor

E_{batt} Energy stored in each battery

E_{string} Energy stored in a single string

$F_{i\to j}$ View factor

h_{con} Conductivity $[\mathrm{W\,m^{-2}\,K^{-1}}]$

I_{d} Inherent degradation factor

$N_{\text{batt,p}}$ Number of battery cells in parallel

$N_{\text{batt,s}}$ Number of battery cells in series

N_{batt} Number of batteries

N_{cell} Total number of solar array cells

N_{par} Number of solar array cells in parallel

N_{ser} Number of solar array cells in series

P_0 Solar Radiation

P_{a} Power produced by the solar arrays

P_{BOL} Specific power output at begin of life

P_{EOL} Specific power output at end of life

P_{e} Power requested by the loads in eclipse

P_{s} Power requested by the loads in sunlight

Q_{albedo} Albedo radiation [W]

Q_{int} Internally generated heat [W]

Q_{ir} Infrared radiation [W]

Q_{sun} Solar radiation [W]

T_{e} Eclipse time

T_{orbit} Period of the orbit

T_{s} Sunlit time

$T_{\mathrm{sc}}^{\mathrm{cold}}$ Temperature in cold case scenario [K]

$T_{\mathrm{sc}}^{\mathrm{hot}}$ Temperature in hot case scenario [K]

T_{space} Deep space temperature [K]

V_{cell} Solar cell voltage

V_{sys} System voltage

X_{e} Lines efficiency in eclipse

X_{s} Lines efficiency in sunlight

$years$ Expected lifetime in years

Acronyms

ACDM ALADIN Control & Data Management. 46, 47, 50, 54

ADCS Attitude Determination and Control Subsystem. 18

ADM Atmospheric Dynamics Mission. 2, 11, 13, 17–19, 21, 22, 24–28, 32–34, 39–42, 44, 46–48, 52–54, 59

AKE Absolute Knowledge Error. 27

ALADIN Atmospheric Laser Doppler Instrument. 2, 5, 6, 20, 24–27, 29, 32–35, 39, 41, 42, 46–49, 52–55

AMR Anisotropic Magnetoresitance. 25

AOCS Attitude and Orbit Control Subsystem. i, 23–27, 32, 35, 41, 46, 48, 49, 57–59

APE Absolute Pointing Error. 27

APSW Application software. 47, 48

ASIC Application-Specific Integrated Circuit. 25

AST Autonomous Star Tracker. 24, 26, 27, 30, 32, 46, 47, 49, 55, 57, 58

AU Astronomical Unit. 39

AVUM Attitude and Vernier Upper Module. 52

BER Bit Error Ratio. 18, 22

BPSK Binary Phase Shift Keying. 18

BR Regulated Bus. 41

BU Unregulated Bus. 41

C/A Coarse Acquisition. 25

CAL Calibration Mode. 41–44

CCD Charge-Coupled Device. 24, 34

CDMU Control and data management unit. 34, 46–48, 50, 54

CESS Coarse Earth Sun Sensor. 24, 26, 27, 30, 32, 52, 53, 55, 57, 58

CFRP Carbon Fiber Reinforced Polymer. 53

CME Coefficient of Moisture Expansion. 53

COM Communication Mode. 35, 41

CP Cold Plate. 33, 37

cPCDU Constellation Power Conditioning and Distribution Unit. 40

CPDU Command Pulse Distribution Unit. 47

CTE Coefficient of Thermal Expansion. 53

DET Direct Energy Transfer. 40

DFU Detection Front-end Unit. 33, 34, 49

DMM Design Maturity Margin. 59

DWL Doppler Wind Lidar. 2, 48

EOL End Of Life. 35, 39, 41

EPS Electric Power Subsystem. ii, 32, 35, 38, 39, 41, 42, 44, 50, 57–59

ESA European Space Agency. 53

FCL Foldback Current Limiter. 40

FDIR Failure Detection, Isolation and Recovery. 19, 47, 48

FDV Flow Diverter Valve. 12

FEM Finite Element Method. 52

FOG Fiber-Optic Gyroscope. 24

FOS Flight Operation Segment. 18

FOV Field of View. 27, 55

FVV Fill and Vent Valve. 12

GPS Global Positioning System. 25, 26, 30, 32, 48, 55, 57, 58

HP Heat Pipes. 33, 37

HPBW Half-power Beamwidth. 21, 22

HSUART High Speed Universal Asynchronous Receiver Transmitter. 46, 47

HT Higher Troposphere. 6

I/O Input/Output. 47

IAM Initial Acquisition Mode. 26, 30

IC Instrument Calibration. 13

ICB Internal Control Bus. 46

ICS In-Situ Cleaning System. 54

IMU Inertial Measurement Unit. 24, 26, 27, 30, 32, 49, 54, 57, 58

IR Infrared. 33

LCL Latching Current Limiter. 40

LCS Laser Cooling System. 33

LEO Low Earth Orbit. 7, 9, 25, 28, 33, 35, 39, 40, 46, 59

LEOP Launch and Early Operations. 4, 5, 13, 18, 20, 32, 35, 40–44

Lidar Light Detection And Ranging. 2

LOS Line of Sight. 2, 7, 24–27, 54

LT Lower Troposphere. 6

LV Latch Valve. 12

MEMS Micro Electro-Mechanical System. 24

MHRM Multipurpose Hold-down and Release Mechanism. 53

MLI Multi Layer Insulator. 33, 37, 58

MMU Mass Memory Unit. 47

MPPT Maximum Power Point Tracking. 40

MTL Mission Timeline. 48

MTM Magnetometer. 25–27, 30, 32, 54, 57, 58

MTQ Magnetorquers. 25, 26, 30, 32, 48, 49, 55, 57, 58

NM Normal Mode. 26, 30

NWP Numerical Weather Prediction. 2, 5–7

OBDH On-Board Data Handling. ii, 32, 35, 41, 45–48, 50, 57–59

OPS Orbit Position Schedule. 48

OSR Optical Solar Reflector. 33, 34, 37, 40, 58

P/L Payload. 32, 35, 41, 49, 50, 55, 57, 58

PCDU Power Control and Distribution Unit. 32, 33, 40, 41, 44, 46, 50, 54, 57–59

PDEC Packet Decoder. 47

PID Proportional-Integral-Derivative. 34

PLH Power Laser Head. 33, 34, 49

PM Processor Module. 46, 47

PMD Propellant Management Device. 11

PROM Programmable Read-Only Memory. 46

PS Propulsion Subsystem. 32, 35, 41, 49, 57, 58

PT Pressure Transducer. 12

PVA PhotoVoltaic Assembly. 57–59

QPSK Quadrature Phase Shift Keying. 18

RAM Random Access Memory. 46

RCS Reaction Control System. 25–27, 30

RF Radio Frequency. 25

RLH Reference Laser Head. 33, 34, 49

RM Reconfiguration Module. 47, 48

RMU Rate Measurement Unit. 24, 26, 27, 30, 32, 54, 57, 58

ROM Read-Only Memory. 47

RTG Radioisotope Thermoelectric Generator. 39

RW Reaction wheel. 12, 13, 49

RWU Reaction Wheel Unit. 25, 26, 30, 32, 54, 57, 58

S/S Subsystem. 32, 35, 41, 49, 50, 57, 58

S3R Sequential Shunt Switching Regulator. 40, 50

SA Solar Array. ii, 38–40, 42–44, 50, 53, 55

SARM Solar Array Rotation Mechanism. 39, 52, 53

SBM Standby Mode. 26, 30

SCM Science Mode. 35, 41

SGM Safe Guard Memory. 47

SK Station keeping. i, 8–10

SM Safe Mode. 26, 27, 30, 35, 36, 41

SMAD Space Mission Analysis and Design [24]. 32

SNR Signal to Noise Ratio. 55

SPENVIS Space Environment Information System. 46, 47

SRP Solar Radiation Pressure. 28, 29

SSO Sun Synchronous Orbit. 7

STR Structure. 57, 58

Strat Stratosphere. 6

SvalSat Svalbard Satellite Station. 19

TC Telecommand and Control. 47

TCM Thruster Control Mode. 26, 27, 30, 35, 41

TCS Thermal Control Subsystem. ii, 31, 32, 34, 35, 37, 41, 50, 57–59

TEC Thermo Electric Cooler. 34, 37

TLE Transmitter Laser Electronics. 49

TME Telemetry Encoder. 47

TP Testing Port. 12

TTMTC Tracking, Telemetry and Telecommand. i, 16, 17, 20–22, 27, 32, 35, 40, 41, 48, 49, 57–59

TTR Telemetry, Telecommand and Reconfiguration. 47, 48

TWTA Travelling Wave Tube Amplifier. 18

WCRP World Climate Research Program. 2

WMO World Meteorological Organization. 6

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